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Numerical evaluation of a ship's manoeuvrability and course keeping control under various wave conditions using CFD

Daejeong Kim *, Soonseok Song, Byongug Jeong, Tahsin Tezdogan

Department of Naval Architecture, Ocean and Marine Engineering, University of Strathclyde, 100 Montrose Street, Glasgow, G4 0LZ, UK

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ABSTRACT

It is critical to estimate the manoeuvring behaviours of ships in waves since it is closely associated with navigation safety at sea. This study aims to predict the wavelength effects on the ship manoeuvring behaviours using a Computational Fluid Dynamics (CFD) based technique which enables resolving the complex interactions between hull, rudder, and propeller in waves. An Unsteady Reynolds-Averaged Navier-Stokes (URANS) model of the well-known benchmarking KRISO Container Ship (KCS) was developed using the dynamic overset grid technique to simulate the ship motions and rudder deflection. To represent the effects of the propeller, the body force method using an infinite-blade actuator disk was used. The manoeuvring analyses were carried out in bow waves of wavelength-to-ship-length ratios varying between 0.7 and 2.0 for constant wave height. Two representative free-running manoeuvres were conducted, namely, course keeping control, and standard turning circle manoeuvres. The analyses revealed that the wavelength has a significant influence on the manoeuvring performance, especially for the approach speeds, ship motions, and manoeuvring indices. The key findings of this research will increase the understanding of ship manoeuvrability in waves and therefore help to enhance navigation safety at sea.

1. Introduction

According to a recent investigation, navigational casualties from collision and grounding incidents account for more than 40% of all marine incidents (EMSA, 2020). Such accidents undoubtedly result from inappropriate manoeuvring actions by navigation officers. A study by Zhang and Li (2017) states that a large number of marine casualties are closely associated with rough sea conditions. Thus, it is of great importance to understand a ship's manoeuvring performance in a real seaway in order to ensure navigational safety at sea.

The influence of waves has been known so significant on the manoeuvrability of a ship as causing the high frequency wave-induced motions (Wang and Wan, 2018). From the hydrodynamics perspectives, such an iterative pattern on ship motions may highly lead to substantial changes in ship's manoeuvring performance. Meanwhile, the wave frequency in deep water only depends on the wavelength based on the linear wave theory. That implies the wavelength is the key parameter to determine ship's manoeuvrability. It has been noted in real operations that a ship can encounter various wave conditions. According to a study by Toffoli and Bitner-Gregersen (2017), wavelengths were estimated to range from about 1.5 m up to approximately 900 m in a fully developed wind sea. In this regard, it seems necessary to investigate a ship's manoeuvrability under various wavelength conditions, which can provide a proper understanding of the relationship between wavelengths and ship manoeuvring performances in a real seaway.

Several approaches have been used to estimate a ship's manoeuvrability in waves: theoretical methods, experimental methods, and Computational Fluid Dynamics (CFD). For theoretical approaches, mathematical models combined with the potential flow theory have been employed for the prediction of ship manoeuvring characteristics in waves. Skejic and Faltinsen (2008) developed a unified model of seakeeping and manoeuvring as a useful tool to predict the manoeuvring behaviours of a ship in regular waves. The manoeuvring performance was predicted by means of the nonlinear slender-body theory, while second-order wave loads were estimated by the potential theory. Subsequently, a series of past research applied similar approaches to estimate the behaviour of a ship motion during the manoeuvre in waves (Seo and Kim, 2011; Subramanian and Beck, 2015; Zhang and Zou, 2016; Paroka et al., 2017). However, their numerical results were commonly found to have discrepancies with the available experimental

* Corresponding author.
E-mail address: daejeong.kim@strath.ac.uk (D. Kim).

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data to some extent. Those studies revealed the shortfalls of those mathematical models in terms of accurately evaluating the ship manoeuvring performance in waves. In fact, those simplified models were not capable of precisely capturing the complex fluid-structure interactions between the hull, rudder, propeller, and waves. Moreover, the potential flow theory was not able to incorporate viscous effects, which diminish the calculation accuracy as stated in Mofidi and Carrica (2014).

On the other hand, experimental approaches have been regarded as the most reliable method to predict a ship's manoeuvring characteristics. The Planar Motion Mechanism (PMM) or free-running tests have played a pivotal role in experimental-based research. However, most of those experiments were conducted in calm water conditions. A large wave basin, manoeuvring control units, and other measurement devices should be additionally fitted to test facilities in order to perform free-running tests in waves. Due to high cost, most research organisations had to convince themselves for restricting their experiments to calm water cases only. On the other hand, some research institutes with state-of-the-art technologies extensively investigated manoeuvring behaviours of ships in waves. To give an example, free-running tests of the ONR Tumblehome model in regular waves were conducted at the Iowa Institute of Hydraulic Research (IIHR) wave basin (Sanada et al., 2013, 2019; Elshiekh, 2014). In addition, free-running tests for the KCS in regular waves were conducted at the Hiroshima University (HR) wave basin (Yasukawa et al., 2021).

With the great progress in computer science and technology, the CFD has become a useful tool to assess a ship's manoeuvrability. Thanks to high accuracy whereas low cost, this approach is now highly preferred by researchers. In particular, a CFD method can incorporate both viscous and rotational effects in the flow and free surface waves, and therefore accurately resolve the complex fluid-structure interactions. Likewise, the initial applications of the direct CFD approach were made under calm water conditions (Mofidi and Carrica, 2014; Broglia et al., 2015; Shen et al., 2015; Wang et al., 2016). The research in this area demonstrated the excellence of the direct CFD method to estimate a ship's manoeuvrability in calm water by comparing CFD results with the experimental data available. Also, there have been several attempts with the direct CFD method to predict manoeuvring behaviours of a ship in wave conditions (Wang et al., 2017, 2018; Wang and Wan, 2018). However, these past studies focused on case-specific analyses with the ONR Tumblehome ship so they are not able to suggest the general patterns or observations on the relation between wave effects and various ship types and geometries.

In fact, many researchers have begun to investigate a ship's manoeuvrability in waves due to an increasing demand in understanding the ship's manoeuvre against various wave effects (IMO, 2014; ITTC, 2017). Nevertheless, due to its brevity, a few remarkable studies managed to discuss the impact of wavelengths on the manoeuvring performance of particular ships. To give some examples, Kim et al. (2019) conducted free-running experiments to examine manoeuvring behaviours of the KRISO Very Large Crude-oil Carrier 2 (KVLCC2) in accordance with various wavelengths. Sanada et al. (2019) examined the manoeuvring performance of the Office of Naval Research Tumblehome (ONRT) surface combatant at different wavelengths. These studies jointly show that the effects of wavelengths on a ship's manoeuvrability largely differ depending on the ship types. For the KVLCC2 in waves, the manoeuvring characteristics (turning trajectories and critical manoeuvring quantities) were found to vary significantly with the wavelength. On the other hand, the variation of such manoeuvring characteristics of the ONRT with respect to wavelengths was revealed much smaller than the KVLCC2. Although being informative for confirming the relation between wavelengths and ship manoeuvring behaviours, the past research studies were not able to offer detailed insight into the understanding on manoeuvring performances of various commercial vessels against different wavelengths. It is because the scope of the past research was limited with some specific benchmarking models, i.e. ONR Tumblehome and KVLCC2. Thus, in order to draw more meaningful information for general application, it is highly believed that the manoeuvrability of other ship types under different wavelength conditions should also be further investigated.

In this context, this paper was motivated to extend the knowledge of wave effects on the manoeuvrability of the KRISO Container Ship (KCS) model; which is the benchmark ship hull of a container ship but not investigated thoroughly in the past research. This study aims to provide a better understanding of a ship's manoeuvrability under real sea states.

In this work the direct CFD method was adopted based on Unsteady Reynolds Averaged Navier-Stokes (URANS) using a commercial CFD software package. Since course-keeping capabilities can be clearly identified in oblique seas (Kim et al., 2021), five bow waves at various wavelengths were modelled in this study. During manoeuvring simulations, the approach speeds were firstly achieved by performing self-propulsion computations. Following this, the course-keeping manoeuvres were carried out to assess the course-keeping qualities in different wavelength conditions. Finally, the turning abilities in waves were examined by carrying out the standard turning circle tests.

2. Methodology

This section will outline the research methodology developed in this paper, with details provided in the contained sub-sections. Fig. 1 shows the methodology overview for the free running CFD simulations. This methodology was organised with four key steps: *1) goal and scope, 2) numerical modelling, 3) execution of free running simulations, and 4) results*. The first step is proposed to establish the overall research aim and the scope of the analysis. In the second step, the numerical modelling is performed to formulate the direct CFD model for free running simulations in different conditions. In the third step, the free running manoeuvres such as course keeping control and turning circle tests are conducted to evaluate the ship's manoeuvrability. In the fourth step, the critical manoeuvring results are combined and plotted in form of graphs which provide a general relationship between the ship's manoeuvring performance and wavelengths.

2.1. Step 1: goal and scope

As stated previously, this paper deals with the effects of wavelengths on a ship's manoeuvring performance in waves. Although ship handling characteristics can be broken into several parts, this study mainly focuses on the course keeping ability and turning ability in different wavelengths. Regarding the scope of the analysis, the ship type and environmental conditions were determined as follows.

In this study all the manoeuvring simulations were carried out for the KRISO Container Ship (KCS) model which is one of the benchmark hull forms. The KCS geometry was appended with an actuator disk and a single rudder as shown in Fig. 2. The main particulars of the model-scale KCS are listed in Table 1 (SIMMAN, 2020).

Table 2 and Fig. 3 jointly illustrate six different conditions adopted in this study for the free-running simulation. The calm water case was carried out to investigate the manoeuvring characteristic of the KCS without any external disturbances. For the wave conditions, the effects of the incident regular waves with wavelength to ship length ratios λ/L_{BP} = 0.7, 1.0, 1.3, 1.6, and 2.0 were examined during ship manoeuvres (the wave height H = 0.048 m and wave encounter angle μ = 225° are constant for all the cases). As previously mentioned, oblique waves were identified as critical conditions significantly affecting the course-keeping control of a ship (Kim et al., 2021). The bow waves (μ = 225°) were therefore chosen in this work to examine the selected cases. The encounter frequency of the wave, ω_e, is calculated as $\omega_{e=}\omega[1 - (\omega U\cos\mu)/g]$, where ω is the wave frequency, U is the ship forward speed, μ is the ship's heading angle relative to the wave direction, and g denotes the gravitational acceleration. It is also of note that the CFD analyses were carried out using deep water conditions.

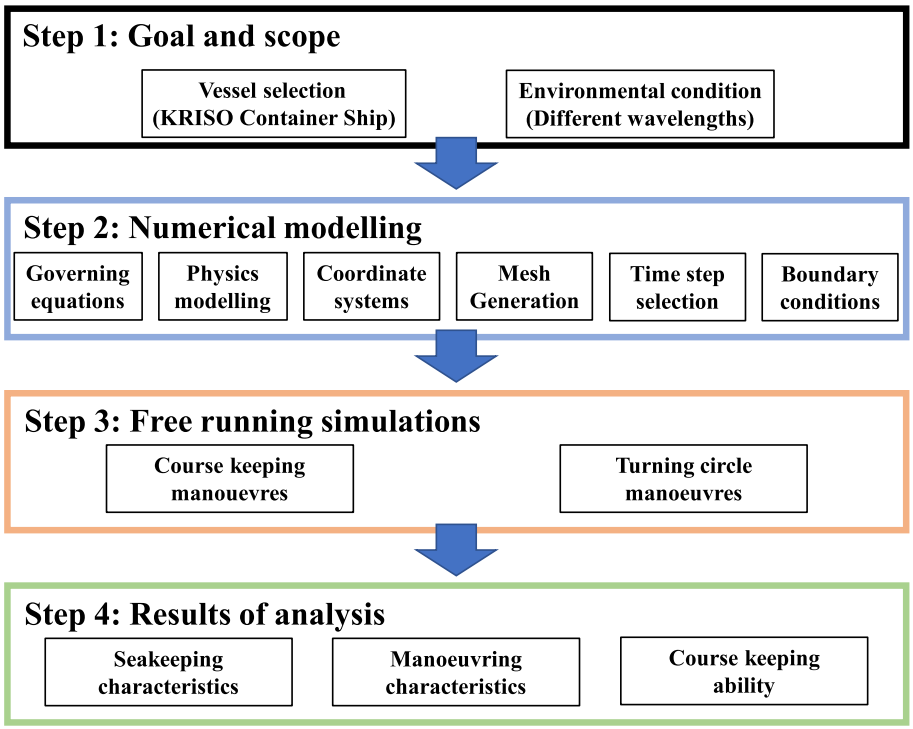


Fig. 1. Methodology overview applied to the direct CFD manoeuvring simulations.



Fig. 2. Geometry of KCS hull with an actuator disk and a rudder.

Table 1
Principal particulars of the KCS.

Main particulars	Symbols	Model scale (1:75.24)
Length between the perpendiculars	L_{BP} (m)	3.057
Length of waterline	L_{WL} (m)	3.0901
Beam at waterline	B_{WL} (m)	0.4280
Draft	D (m)	0.1435
Displacement	Δ (m^3)	0.1222
Block coefficient	C_B	0.651
Ship wetted area with redder	S (m^2)	1.6834
Longitudinal centre of buoyance	% L_{BP}, fwd +	−1.48
The metacentric height	GM (m)	0.008
Radius of gyration	K_{xx}/B	0.49
Radius of gyration	K_{yy}/L_{BP}, K_{zz}/L_{BP}	0.25
Propeller diameter	D_P (m)	0.105
Propeller rotation direction (view from stern)		Right hand side
Rudder turn rate	(deg./s)	20.1

2.2. Step 2: numerical modelling

In this study the free-running simulations of the KCS were conducted using the commercial CFD software STAR-CCM+, version 15.04. This chapter provides details of the numerical methods applied in this paper.

2.2.1. Governing equations

In order to solve the governing equations, an Unsteady Reynolds-Averaged Navier-Stokes (URANS) approach was used in this study. The mass and momentum conservation equations were solved by employing the commercial CFD software. The averaged continuity and momentum equations for unsteady incompressible flows are given in tensor notation and Cartesian coordinates as follows (Ferziger et al., 2020):

$$\frac{\partial\left(\rho\overline{u}_i\right)}{\partial x_i}=0 \tag{1}$$

$$\frac{\partial\left(\rho\overline{u}_i\right)}{\partial t}+\frac{\partial}{\partial x_i}\left(\rho\overline{u}_i\overline{u}_j+\rho\overline{u_i'u_j'}\right)=-\frac{\partial\overline{p}}{\partial x_i}+\frac{\partial\overline{\tau}_{ij}}{\partial x_i} \tag{2}$$

where ρ indicates the fluid density, $\overline{u}_i$ indicates the averaged velocity vector, x_i (i = 1, 2, 3) are the Cartesian coordinates, $\rho\overline{u_i'u_j'}$ is the Reynolds stresses, $\overline{p}$ is the mean pressure and $\overline{\tau}_{ij}$ are the mean viscous stress tensor components. This stress tensor for a Newtonian fluid can be shown in Eq. (3)

$$\overline{\tau}_{ij}=\mu\left(\frac{\partial\overline{u}_i}{\partial x_j}+\frac{\partial\overline{u}_j}{\partial x_i}\right) \tag{3}$$

in which μ means the dynamic viscosity. The RANS method for turbulent flows requires that the Reynolds stresses in Eq. (4) be appropriately modelled. A common method uses the Boussinesq hypothesis.

Table 2
The simulation cases to which the CFD model is applied.

Case no. C	Approach speed U_0 (m/s)	Propeller rev. (RPS)	Wave height H (m)	Encounter Angle μ (degrees)	Encounter Freq. ω_e (rad/s)	Encounter Period T_e (s)	Wave/ship length λ/L_{BP}
1	1.094	13.38	Calm water	- (Calm sea)	–		–
2	0.904	13.38	0.048	225 (Bow sea)	7.267	0.865	0.70
3	0.945	13.38	0.048	225 (Bow sea)	5.863	1.071	1.00
4	1.024	13.38	0.048	225 (Bow sea)	5.088	1.235	1.30
5	1.050	13.38	0.048	225 (Bow sea)	4.506	1.394	1.60
6	1.054	13.38	0.048	225 (Bow sea)	3.943	1.594	2.00

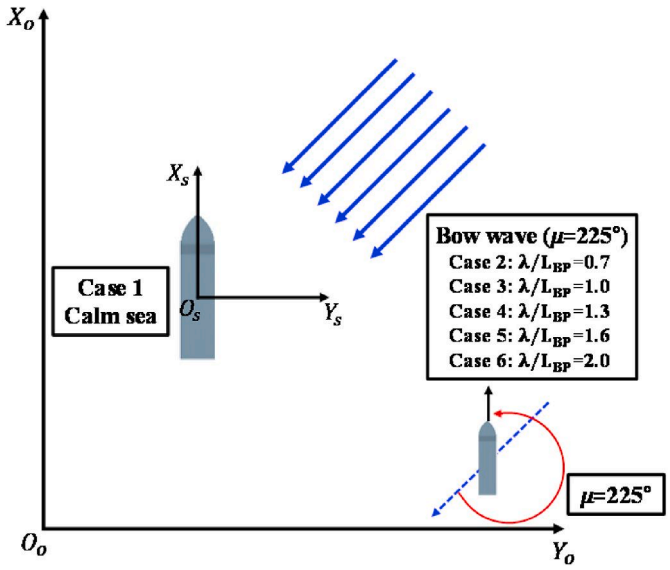


Fig. 3. A schematic view of the simulation cases applied to this study.

$$-\rho\overline{u}_i\overline{u}_j = u_t\left(\frac{\partial\overline{u}_i}{\partial x_j}+\frac{\partial\overline{u}_j}{\partial x_i}\right)-\frac{2}{3}\left(\rho k+u_t\frac{\partial\overline{u}_k}{\partial x_k}\right)\delta_{ij} \tag{4}$$

where the u_t is the turbulent eddy viscosity, and k and δ_{ij} are the turbulence kinetic energy and Kronecker delta, respectively.

To describe the flow of an incompressible viscous fluid numerically, the governing equations are discretised on a fixed Cartesian grid using the finite volume method. A second-order convection scheme was used for the convective terms and implicit Euler time stepping. Additionally, a second-order temporal discretization was used for the time domain solution. The flow equations were solved in a segregated manner. The RANS solver uses a predictor-corrector approach to link the continuity and momentum equations.

2.2.2. Physics modelling

In this numerical study, the Shear Stress Transport (SST) turbulence model (Menter, 1994) was used to represent turbulence effects. The SST k−ω model effectively blends a k-ε model in the far-field with a k−ω model near the wall, which gives highly accurate predictions of boundary layer separation. Menter's SST model has also been employed in several manoeuvring studies (Mofidi and Carrica, 2014; Wang and Wan, 2018; Liu et al., 2020; Kim et al., 2021). The current CFD model used the all-y+ wall treatment, which is a hybrid treatment that emulates the low-y+ wall treatment for fine grids (when the wall y+ < 5), and the high-y+ wall treatment for coarse grids (when the wall y+>30). In this paper, the average value of y+ on the hull surface is higher than 30.

The "Volume of Fluid" (VOF) technique, first introduced by Hirt and Nichols (1981), was applied to model the interface between the two phases (air and water). This VOF method can represent the free surface flow around ships in incident waves. Due to its high numerical efficiency, the model is suited for flow simulations in which each phase constitutes a large structure, with a low overall contact area between phases (CD-Adapco, 2020). This VOF method can be combined with the 6 DOF motion model for ship manoeuvring simulations in waves. Fig. 4 presents a snapshot of free surface elevation with the volume fraction profile on the hull. In the figure, the value of the volume fraction varies from 0 (air) to 1 (water). For example, the water volume fraction value of 0.5 indicates the position of the water-air interface which is filled with 50% water and 50% air. Additionally, the High-Resolution-Interface Capturing (HRIC) scheme is used in VOF simulations to maintain a sharp interface between the incompressible fluid phases.

The Dynamic Fluid Body Interaction (DFBI) method was used to simulate the realistic motion of the ship. For the current CFD model, the DFBI module enables the ship to move with 6 degrees of freedom to estimate the manoeuvring performance of a ship in waves. The 6 DOF motion solver computes fluid forces, moments, and gravitational forces acting on the ship and solves the equations of motion to calculate the new position of the ship. The equations of motion are given as the following expressions:

$$\frac{\mathrm{d}(m\mathbf{V})}{dt}=\mathbf{f} \tag{5}$$

$$\mathrm{M_b}\frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{dt}}+\boldsymbol{\omega}\times\mathrm{M_b}\,\boldsymbol{\omega}=\mathbf{m} \tag{6}$$

where m is the mass of body, $\mathbf{V}$ is the velocity vector of its centre of mass, $\mathbf{f}$ is the resultant force vector acting on the ship, $\mathrm{M_b}$ indicates the tensor of moments of inertia in the body-fixed coordinate system, $\boldsymbol{\omega}$ is the angular velocity vector of the body and $\mathbf{m}$ denotes the resultant moment vector acting on the body.

The body force propeller method using an infinite-blade actuator disk was employed to represent the effects of a propeller. The axial and tangential forces of the modelled propeller and its effects on the flow are calculated using this model. The integration of these forces over the disk gives the thrust and torque of the propeller, which can be then combined with the DFBI module. The propeller effects are treated as body force terms in the governing equations. The computational time to perform free running manoeuvres can be reduced by applying this simplified disk model compared to CFD models using a real propeller geometry. To model an actuator disk, numerous input parameters are required including propeller dimensions, the rotational speed, and performance curve results. In this paper, input variables were selected based on the propeller experimental data conducted by Hiroshima University. It is of note that the propeller-induced velocity was corrected to model the effects of the propeller efficiently according to the induced velocity correction property, stated in Neitzel et al. (2015).

The 5th-order Stokes wave model was adopted to generate regular waves. The model is based on the theory of the 5th-order Stokes wave theory developed by Fenton (1985). This wave model can more closely represent a realistic wave propagation than a wave generated by the first-order Stokes approximation, according to CD-Adapco (2020). The 5th-order Stokes waves were used by several authors in similar studies (Tezdogan et al., 2015; Kim et al., 2021).

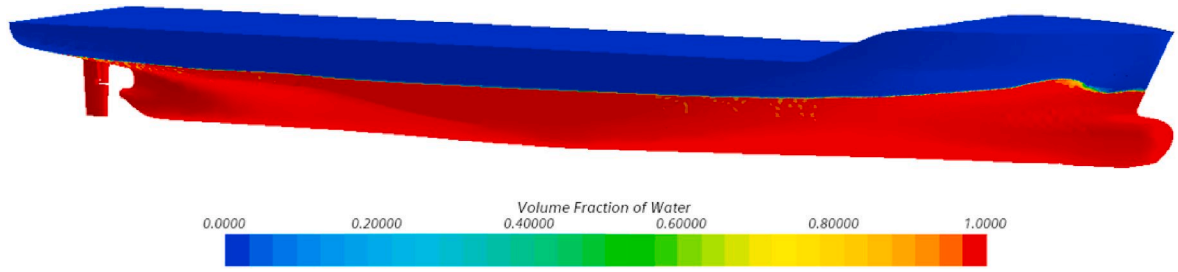


Fig. 4. Representation of free surface on the ship hull with the volume fraction field.

2.2.3. Coordinate systems

For the solution of the fluid flow and ship motions, four right-hand coordinate frames were defined as indicated in Fig. 5: (1) Earth-fixed frame ($O_o - X_oY_oZ_o$), (2) Ship-fixed frame (o_s - $x_sy_sz_s$), 3) Propeller-fixed frame (o_p - $x_py_pz_p$), and (4) Rudder-fixed frame (o_r - $x_ry_rz_r$). The earth-fixed frame is regarded as an inertial coordinate system with both translation and rotation motions not allowed. The origin O_o of the earth-fixed frame can be selected as any point on the surface of the earth. The ship-fixed frame is a moving reference frame which is fixed to the ship. The origin o_s of the ship-fixed frame is positioned at the centre of mass of the ship. The axis o_sx_s is positive towards the bow, the axis o_sy_s is positive towards the starboard, and the axis o_sz_s is positive downwards. The propeller-fixed frame is introduced to apply the body force method with an actuator disk. The thrust produced by the disk acts in the x_p direction of this coordinate system. Finally, the rudder-fixed frame is employed to control the rudder rotation motion. By the rudder movement, the ship manoeuvring tests can be conducted.

2.2.4. Mesh generation

Grid generation was performed by using the built-in meshing facility in STAR-CCM+, which uses the Cartesian cut-cell method. This automatic meshing tool generates the volume or surface grids appropriate for the finite volume method. A trimmed cell mesher was applied to generate the volume grids, which are mainly composed of hexahedral cells with trimmed cells adjacent to the surface. A surface remesher was employed to obtain a high-quality surface mesh. The remesher tool optimises the surface of the geometry for the volume meshing. The resulting surface mesh was made up of rectangular grids on the surface. Six layers of prismatic cells were created to resolve the surface boundary layer by using a prism layer mesh model. The height of the first layer mesh was 2.4×10^{-5} m, which ensures that the hull surface y+ is more than 30. The grid refinement was applied in the free surface region to accurately capture the non-linear characteristics of regular waves. The grids were also refined near the tight gap parts between the rudder blade and horn as well as the propeller wake region. The obtained mesh for the CFD manoeuvring simulations of KCS is presented in Fig. 6.

In order to simulate ship manoeuvres in waves, the generation of free surface grids was conducted based on the guidelines for ship CFD applications from ITTC (2011). According to these recommendations, a minimum of 80 cells wavelength was required on the free surface, while a minimum of 20 cells should be generated in the vertical direction where the free surface is expected. Thus, the free surface refinement was implemented in the whole free surface area (x, y, and z directions, respectively) to ensure stable wave generation in the computational domain. The total grid number for each manoeuvring simulation is shown in Table 3. It should be noted that the total cell numbers differ due to variation in the wavelength conditions, in which at least 80 cells per wavelength were created, as explained above.

The dynamic overset grid approach was applied in this study to resolve the complex manoeuvring behaviour in waves. The conventional mesh approaches, e.g., rigid, and deforming motion mesh, are available to simulate the motion of bodies. However, these methods experience some difficulties in simulating several bodies with large amplitude motions due to cell quality problems. On the other hand, the overset grid

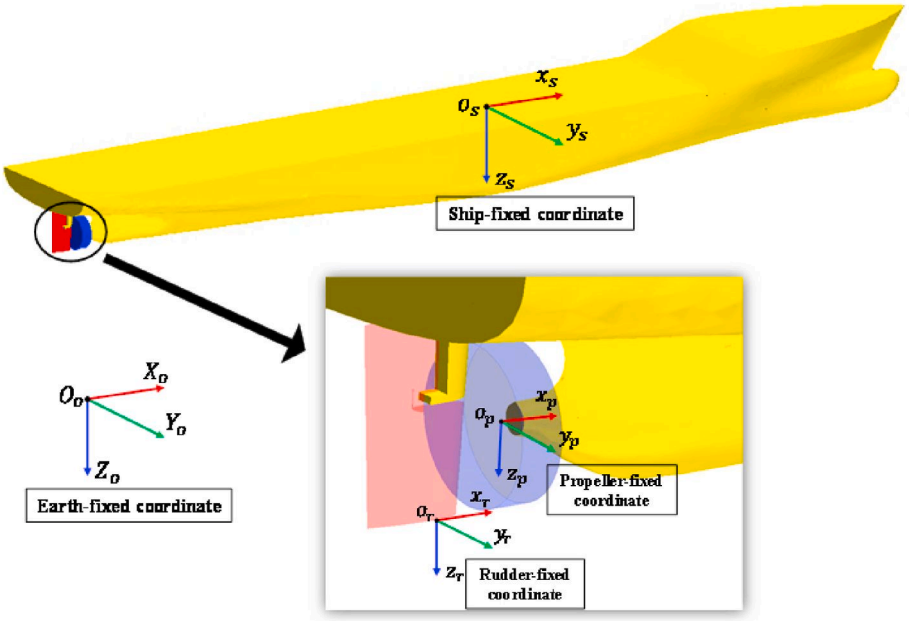
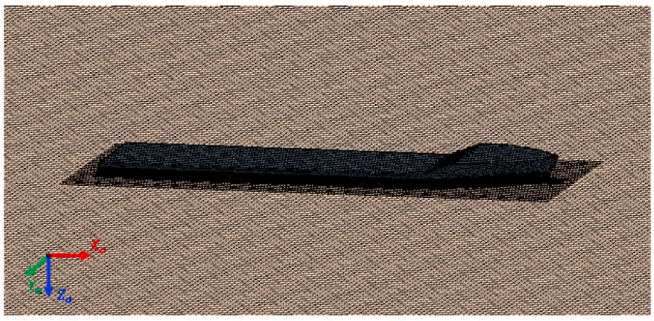
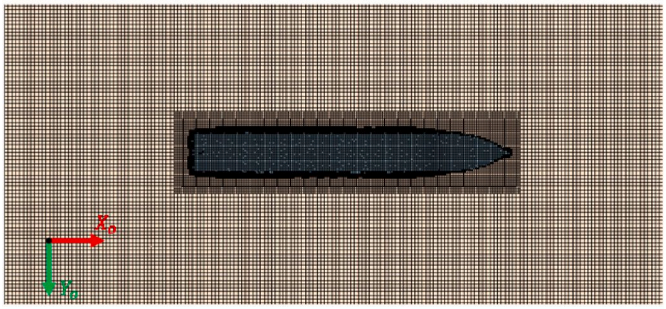


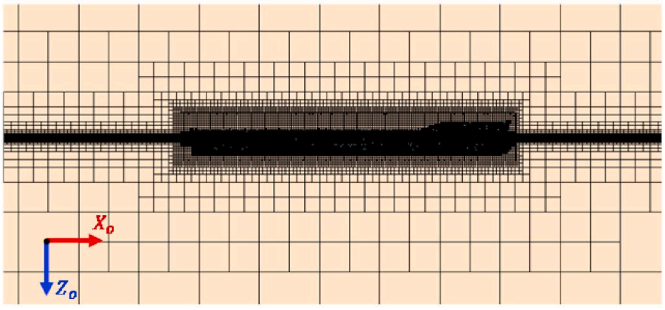
Fig. 5. The Coordinate systems of the ship manoeuvring simulation used in this study.



a) Mesh for the ship and free surface



b) Top view cross-section of the domain



c) Profile view cross-section of the domain

Fig. 6. Mesh structure of the computational domain.

Table 3
The total cell numbers for manoeuvring simulations.

Case no.	Total cell number
1 (Calm water)	5,391,040
2 (Bow wave, $\lambda/L_{BP} = 0.7$)	11,411,380
3 (Bow wave, $\lambda/L_{BP} = 1.0$)	7,189,514
4 (Bow wave, $\lambda/L_{BP} = 1.3$)	6,372,141
5 (Bow wave, $\lambda/L_{BP} = 1.6$)	5,348,493
6 (Bow wave, $\lambda/L_{BP} = 2.0$)	4,776,950

method has distinct advantages compared to the traditional grid methods. The overset regions, which encompass moving bodies, allow large and arbitrary relative motion of bodies while preserving a high-quality mesh. Computational costs can also be saved by generating sufficiently refined grids around important moving parts. The same overlapping method has been employed by several authors in similar studies dealing with the free-running CFD simulations (Mofidi and Carrica, 2014; Shen et al., 2015; Wang et al., 2016; Kim et al., 2021).

In our CFD simulations, three computational regions were generated: 1) background region, 2) overset region around the ship hull, and 3) overset region surrounding the rudder blade, as seen in Fig. 7. The overset regions were created based on the hole-cutting process. The hole-cutting method in STAR-CCM+ automatically couples the overset regions with the background region through the overset interfaces. From the hole-cutting process, four types of cells are generated: 1) active cells, 2) passive cells, 3) donor cells, and 4) acceptor cells. Detailed information on the overset mesh methodology can be found in Zhang et al. (2020). It is of note that creating the overlapping grids around the rudder is quite challenging due to the tight gap between the rudder blade and horn, whereas the overset grid generation around the ship hull is more straightforward.

2.2.5. Determination of the time step

A proper time-step should be selected for the numerical stability of manoeuvring simulations. The Courant number (CFL) is the ratio of physical time step to the mesh convection time scale. The CFL number formula is given as follows:

$$\mathrm{CFL} = \frac{U\Delta t}{\Delta x} \tag{7}$$

where U is the flow velocity, Δt is a representative time step of the simulation and Δx indicates the mesh cell dimension. It is generally equal to 1 or less for stability in numerical solutions.

For implicit unsteady simulations, the time step can be determined by the flow properties, rather than the Courant number. Thus, the time step was chosen based on the recommendation of ITTC (2011) throughout all the simulations. According to the recommendation, at least 100 time steps per encounter period should be used for the wave simulation. Given this, the time step size was chosen to be 0.005s, which satisfies this condition for all the wave cases.

2.2.6. Computational domain and boundary conditions

It is of great importance to determine appropriate boundary conditions for CFD simulations. Several combinations of boundary conditions can be applied to resolve CFD problems. However, the determination of the most proper boundary conditions can provide a good approximation of the problem, and thus achieve a reliable solution.

Fig. 8 describes the computational domain and boundary conditions of the present CFD model with the dimensions of the domain. The two opposite faces at the x-direction of the domain, i.e., the upstream and downstream boundaries, were set as a velocity inlet condition. The side walls (the two opposite faces at the y-direction of the domain) and bottom wall were also modelled as a velocity inlet. It prevents a velocity gradient from occurring between the fluid and the wall. A pressure outlet condition was applied to the top boundary. No-slip wall boundary conditions were applied to the surface of the hull and rudder.

For the calm water simulation, the VOF wave damping capability of the software was applied to the up-and-down stream boundaries and side walls with a damping length equal to 1.0 L_{BP} (~3.06 m). VOF waves generated by the ship can be damped in the vicinity of selected boundaries by introducing vertical resistance to the vertical motion. Thus, this approach can minimise undesired wave reflection at boundaries.

For the wave simulations, the wave forcing capability was used at all the vertical boundaries to reduce the size of the computational domain without compromising the solution's fidelity. The solution of the discretised Navier-Stokes equations is forced towards the theoretical fifth-order Stokes wave solution (Fenton, 1985) over a forcing length equal to 1.0 L_{BP} (~3.06 m) from the boundaries. Ferziger et al. (2020) point out that the forcing method can minimise disturbances generated by the presence of the vessel in the forcing zone, which helps to avoid wave reflections from the boundaries. The forcing zone applied in this study is illustrated in Fig. 9, in which relative zone distance means the relative distance between a cell and the nearest boundary of the forcing zone.

Different motions were assigned to each computational region for the free-running simulations. The ship hull overlapping region was allowed

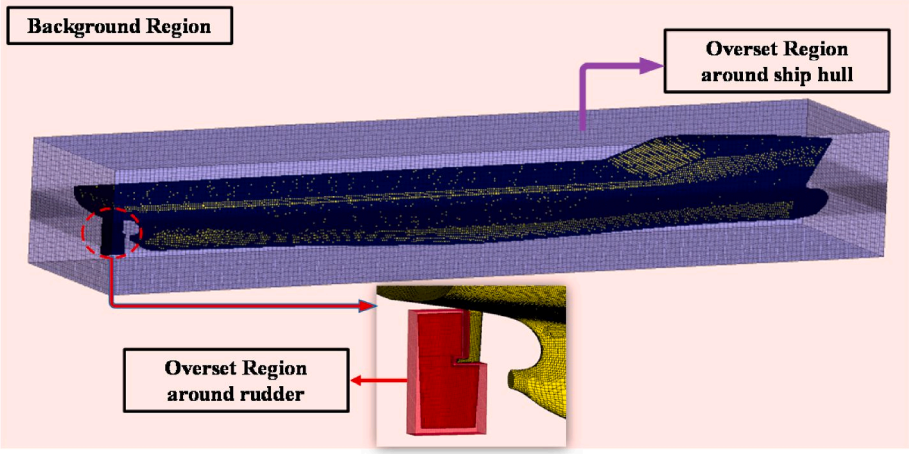


Fig. 7. Overset mesh configuration of the domain.

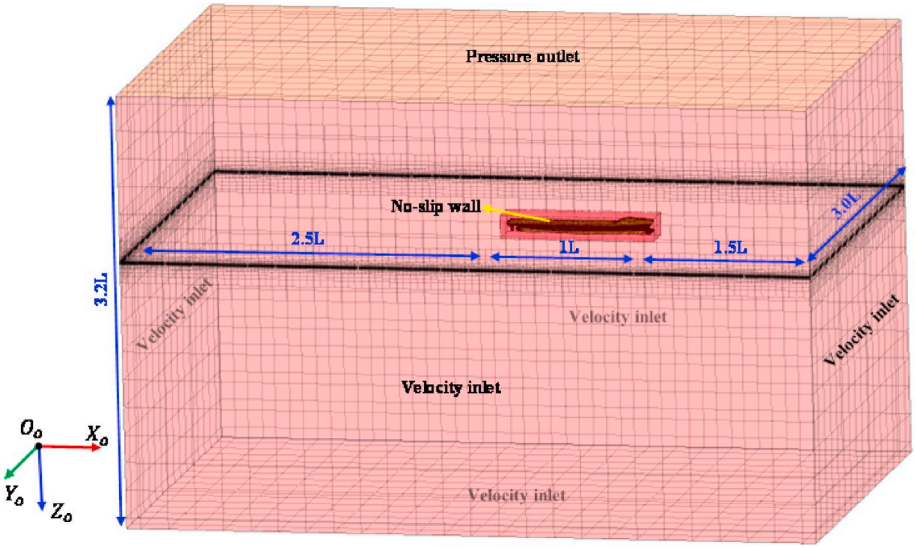


Fig. 8. Computational domain and the boundary conditions (L: Ship length).

to move with six degrees of freedom by considering all forces and moments acting on the ship, including the effects of the actuator disk. The rudder overset region, which encompasses the rudder blade, was compelled to move with the hull. An additional rotating motion around its shaft was imposed to perform the ship manoeuvre. The background region was forced to move with 3 DOF by imposing the same velocities of the ship surge, sway, and yaw motions, such that the relative static motion between the ship and the background region was achieved. It should be noted that the application of other motions, i.e., heave, pitch, and roll may adversely affect the free surface resolution, which cannot guarantee the desired wave propagation. The applied domain motions are illustrated in Fig. 10.

2.3. Step 3: free running simulations

2.3.1. Course keeping simulation

The self-propulsion computation should be initially performed to achieve the approach speed before conducting the course keeping and turning circle manoeuvres. The course-keeping control of a ship has become an important concern in the field of the ship manoeuvre since it is closely related to the economy and the safety of ship operations at sea. A ship operation in a real sea environment is inherently under the influence of external disturbances such as waves and winds. The course-keeping ability can be adversely affected in the presence of these significant disturbances. In order to evaluate a ship's steering ability, in this paper, a Proportional-Integral-Derivative (PID) controller was applied. During course keeping manoeuvres, the ship was free to move in six degrees of freedom (6DOF) with the moving rudder controlled by the PID feedback controller to maintain the ship's course. The revolutions per second (RPS) of the actuator disk was set to be 13.38 n/sec, which is identical to the propeller speed used for the KCS manoeuvring analyses in waves by Kim et al. (2021). The rudder angle is controlled by the feedback controller to enable the ship to sail straight in waves. The control function is defined by the following expressions:

$$\delta(t) = K_p e(t) + K_i \int_0^t e(t)dt + K_d \frac{de(t)}{dt} \tag{8}$$

$$e(t) = \psi(t) - \psi_c \tag{9}$$

where, $\delta(t)$ indicates the rudder angle, $\psi(t)$ indicates the instantaneous yaw angle at a given time, ψ_c indicates the target yaw angle which was set to be 0° to make the ship advance straight. K_p, K_i, and K_d. mean the proportional, integral, and derivative control gains, respectively. It is quite challenging to determine optimal values of the control gains for

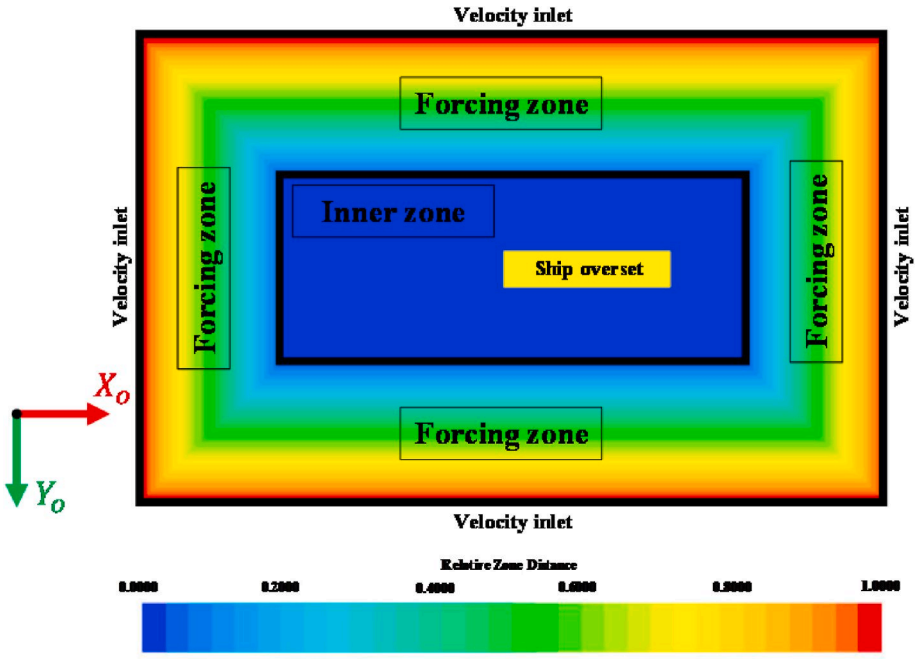


Fig. 9. A schematic view of the forcing zone applied to the CFD model.

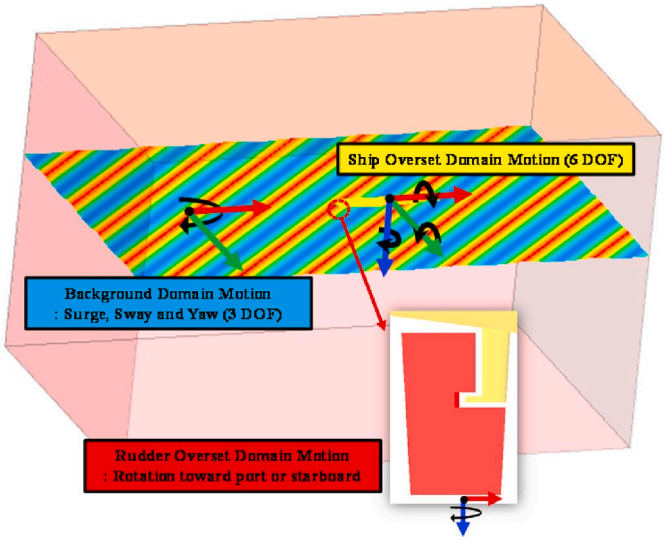


Fig. 10. The view of the motions of the created domains.

the achievement of good steering ability. In this work, the gains were set to $K_p = 5$, $K_i = 0.05$, *and* $K_d = 3$after several trials and errors. It is of note that feedback control gains of the autopilot can be changed depending on environmental conditions.

2.3.2. Turning circle simulation

In order to evaluate the turning ability of the ship, the standard turning circle test was conducted. For the standard turning circle test, the rudder angle $\delta(t)$ is executed by the following module:

$$\delta(t) = \begin{cases} \max(0,\ kt),\ \delta \le 35 \\ 35 \end{cases} \tag{10}$$

where k is the maximum rudder rotational rate. The maximum rudder rate is set to $k = 20.1°/s$ corresponding to $2.32°/s$ on full scale according to the experimental data. From the definitions above, the rudder angle is turned to 35° and kept constant until the vessel completes a turning circle test.

2.4. Step 4: results of analysis

The step 4 was to estimate the manoeuvring behaviour of the KCS in different wavelength conditions. By applying a wide range of wavelengths in the ship manoeuvre, the effects of the wavelength on the ship's manoeuvrability were examined. The time to complete the computation was around 336 wall clock hours and 13440 CPU hours with 40 CPU processors for each manoeuvring simulation.

3. Case studies (results of step 4)

3.1. Verification and validation study

In this work a verification study was not conducted for free running ship manoeuvres in waves, due to the high computational cost involved in manoeuvring problems. In general, the process of refining and coarsening the overset parts is highly demanding, as discussed in Carrica et al. (2007). Because of this reason, a verification study has not been performed for free running simulations by most researchers (Mofidi and Carrica, 2014; Wang and Wan, 2018; Wang et al., 2018; Liu et al., 2020). However, Kim et al. (2021) conducted spatial and temporal convergence studies of the same direct CFD model applied to this work. According to Kim et al. (2021), numerical uncertainties for critical manoeuvring parameters were found to be a maximum of 0.28% in the spatial convergence test (Table 4) and 0.19% in temporal convergence study (Table 5), based on Grid Convergence Index (GCI) approach.

Before performing the free running simulations, the open-water performance of the actuator disk applied to the current CFD model was initially estimated and compared with the experimental data from Hiroshima University. Fig. 11 presents this study's CFD results with comparison to the experiments. In figure J represents the advance ratio, K_T is the thrust coefficient, K_Q is the torque coefficient, and η_0 is the efficiency of the propeller. The obtained results show good agreement with the experiments, which indicates the applied actuator disk is capable of predicting the propeller performance in terms of thrust and torque.

Unfortunately, experimental results for the ship manoeuvre in bow

Table 4
Calculation of the discretization error for the spatial convergence study, key variables: Advance, transfer, and tactical diameter, adapted from Kim et al. (2021).

	Advance	Transfer	Tactical diameter
N_1	7,189,514	7,189,514	7,189,514
N_2	4,825,180	4,825,180	4,825,180
N_3	3,153,041	3,153,041	3,153,041
r_{21}	1.14	1.14	1.14
r_{23}	1.15	1.15	1.15
ϕ_1	7.866 m	3.089 m	7.968 m
ϕ_2	7.924 m	3.141 m	8.024 m
ϕ_3	8.598 m	3.662 m	9.131 m
ε_{32}	6.74E-01	5.21E-01	1.11E+00
ε_{21}	5.82E-02	5.18E-02	5.60E-02
s	1	1	1
e_a^{21}	0.74%	1.68%	0.70%
q	−1.69E-01	−1.61E-01	−1.98E-01
p_a	1.72E+01	1.62E+01	2.10E+01
ϕ_{ext}^{21}	7.86E+00	3.08E+00	7.96E+00
e_{ext}^{21}	0.08%	0.22%	0.05%
GCI_{fine}^{21}	0.11%	0.28%	0.06%

Table 5
Calculation of the discretization error for the temporal convergence study, key variables: Advance, Transfer, and Tactical diameter, adapted from Kim et al. (2021).

	Advance	Transfer	Tactical diameter
Δt_1	0.005s	0.005s	0.005s
Δt_2	0.01s	0.01s	0.01s
Δt_3	0.02s	0.02s	0.02s
r_{21}, r_{23}	2	2	2
ϕ_1	7.86582 m	3.089 m	7.968 m
ϕ_2	7.84181 m	3.109 m	7.996 m
ϕ_3	7.76073 m	3.214 m	8.313 m
ε_{32}	−8.11E-02	1.04E-01	3.17E-01
ε_{21}	−2.40E-02	2.02E-02	2.81E-02
e_a^{21}	0.31%	0.65%	0.35%
p_a	1.76E+00	2.37E+00	3.49E+00
ϕ_{ext}^{21}	7.876E+00	3.084E+00	7.965E+00
e_{ext}^{21}	0.128%	0.157%	0.034%
$GCI_{\Delta t_1}^{21}$	0.161%	0.196%	0.043%

waves are not available in the literature, and therefore could not be presented in this paper. However, the current CFD approach (Kim et al., 2021) for free running manoeuvres has been extensively validated against experimental data obtained from Yasukawa et al. (2021) as fully described in Figs. 12 and 13 and Tables 6 and 7. The agreement between the predicted CFD results and the experiments was reasonably good with errors up to 5% in the calm sea and 8% in the head sea, respectively. More detailed information about the comparisons can be found in the study of Kim et al. (2021). This clearly demonstrates that the present CFD model is a reliable tool to estimate the manoeuvring behaviours of a ship in waves of different wavelengths.

3.2. Course keeping control

3.2.1. Approach speed, ship resistance, and ship motions

The time histories of the predicted approach speed, ship resistance, pitch motion, and heave motion under course keeping control for all the cases are shown in Fig. 14. The unsteady time histories of the CFD results were analysed by Fourier Series (FS). The average value of the time

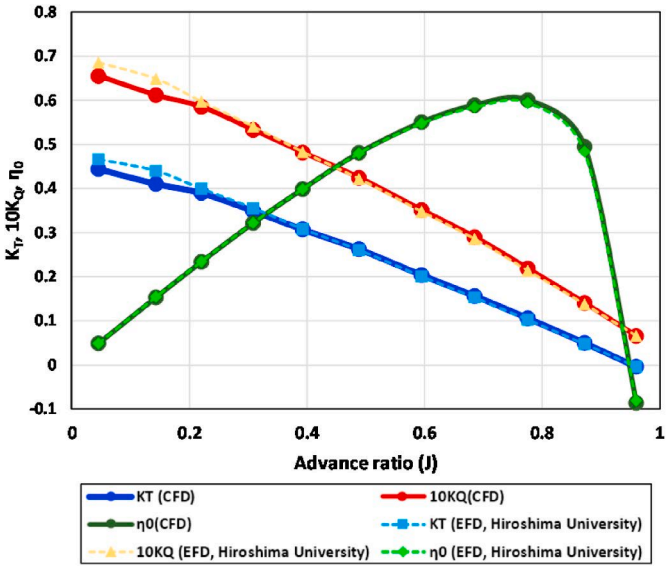


Fig. 11. Propeller open water test results and comparison.

history of the obtained results was quantified by the 0th harmonic FS term, while the mean amplitude of the oscillation of the values was expressed by the 1st harmonic FS term. It is of note that the zeroth and first FS term have been regarded as the fundamental components in the linear system. In this work, the 0th and 1st order terms for the force and motions were calculated, as given in Table 8.

Fig. 14 and Table 8 jointly describe the manoeuvring performance of the KCS under course keeping control for varying wavelengths. It should be borne in mind that regular waves with height of H = 0.048 m and wave encounter angle μ = 225° were applied throughout all the wave simulations, only changing the wavelength. As stated previously, the wavelength to ship length ratio (λ/L_{BP}) ranged from 0.7 to 2.0. Fig. 14 (a) shows a significant change in the approach speed of the KCS when the ship was advancing forward in waves of different wavelengths. The mean approach speed in the calm water was reported to be 1.094 m/s corresponding to 18.45 knots in full scale (Froude number = 0.20). The reduction of the mean forward speed was observed to be 17.4% for the $\lambda/L_{BP} = 0.7$, 13.6% for the $\lambda/L_{BP} = 1.0$, 6.4% for the $\lambda/L_{BP} = 1.3$, 4.0% for the $\lambda/L_{BP} = 1.6$, and 3.7% for the $\lambda/L_{BP} = 2.0$ when compared to the calm water case. This is because the ship resistance increases with the reduction of the wavelength at a given wavelength range ($0.7 < \lambda/L_{BP} < 2.0$). It is of note that the approach speed and ship resistance oscillated considerably in long waves ($\lambda/L_{BP} > 1.0$), but the oscillations significantly decreased when the ship advanced in short waves ($\lambda/L_{BP} = 0.7$).

Fig. 14 (c) and (d) present the comparisons of the ship motion time histories in different wavelength conditions. From this figure, it was observed that the frequency of ship motions in waves was closely related to the wave encountering frequency. Short-term oscillations of the ship motions with regular periodicity were noticed in the shorter waves, whereas relatively long-term oscillations were found in longer waves in a given period. The amplitude of ship motions was also found to vary depending on the wavelength. The wave steepness (H/λ) was identified as the important factor involved in the amplitude of the pitch motion. It should be reminded that the wave height was kept constant for all the wave simulations, which indicates the wave steepness decreased with the increase in the wavelength. The increase of wavelengths (=the decrease of wave steepness) generally led to a decrease in the pitch moment, causing the pitch amplitude to decrease. In other words, the greater the steepness of the wave, the larger the amplitude of the pitch can be. Exceptionally, the largest pitch motion was noticed at the bow waves of $\lambda/L_{BP} = 1.0$ with a mean amplitude of 1.87°. Carrica et al. (2011) state that the maximum pitch response occurs when the

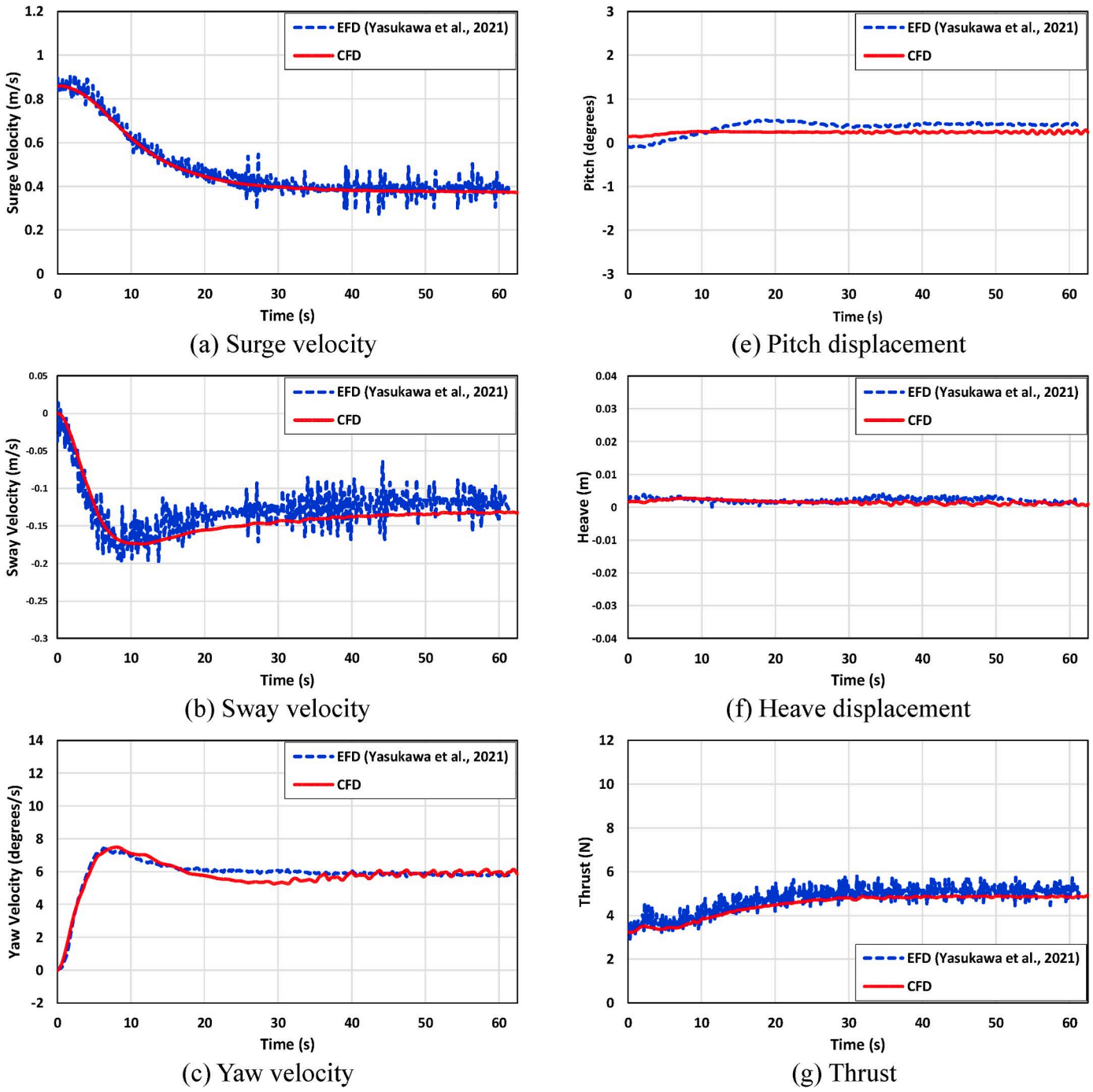


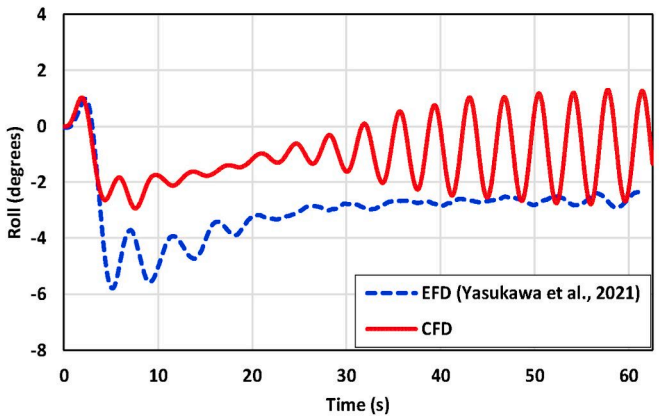
Fig. 12. The comparisons of the time histories of the ship velocities, motions, propeller characteristics, and turning trajectory during turning a circle manoeuvre in calm water, Kim et al. (2021).

encounter frequency (f_e) is equal to the natural pitch frequency (f_n). In this study, the encounter frequency at the bow waves of $\lambda/L_{BP} = 1.0$ was calculated at 0.93hz, which is very close to the natural frequency of the KCS pitching system (0.90hz), as obtained from Carrica et al. (2011). Regarding the heave motion, the largest oscillation amplitude was found in the bow waves of λ/LBP = 2.0 with a mean value of 0.002 m. The amplitude of the heave motion was found to increase as the wavelength increased, which was the opposite trend compared to the pitch motion. It is of note that the amplitude of the vertical motion gets closer to the wave amplitude when the wavelength along the ship length becomes longer (Liu et al., 2020).

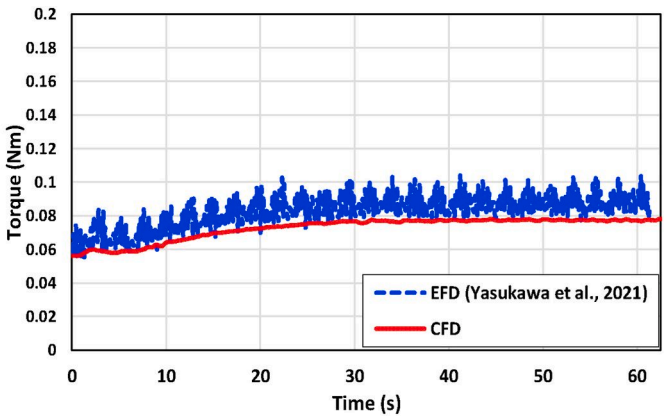
The global wave pattern around the KCS under course keeping control is depicted in Fig. 15 in which the Kelvin waves generated by the advancing ship are clearly visible. It was revealed from the figure that the free surface elevation pattern changed for changing wavelengths.

3.2.2. *Course keeping capabilities*

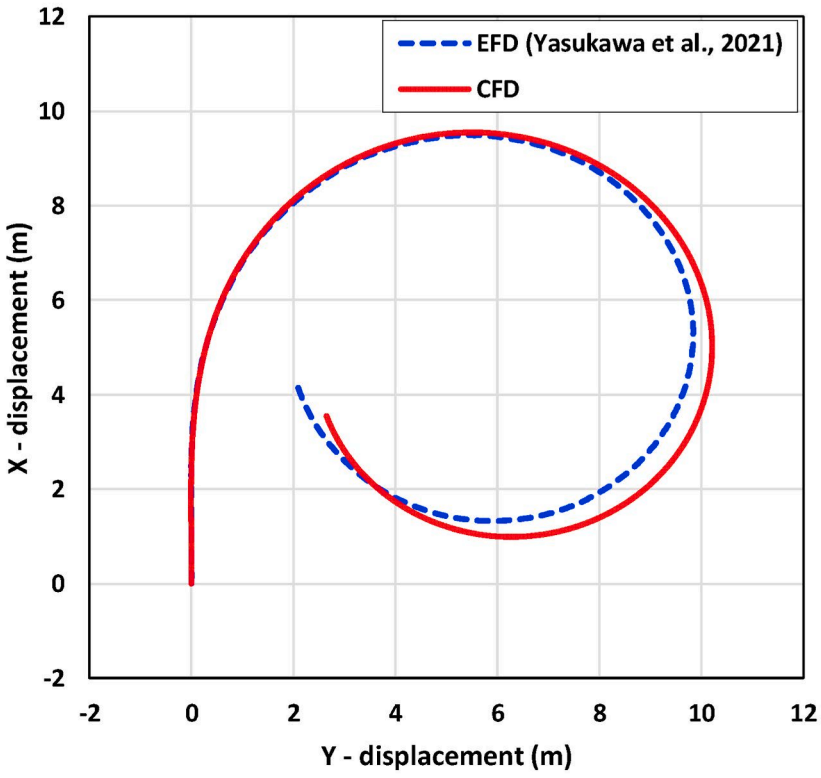
In general, vessels sail to the port of destination according to the planned route which can ensure safe navigation at sea. When a ship advances in a real seaway, a straight-ahead condition will result in some level of yaw, sway, and drift due to the presence of external disturbances. This may cause a deviation from the prescribed route. To



(d) Roll displacement



(h) Torque



(i) Trajectory in calm water

Fig. 12. (*continued*).

minimise the deviation, the ship autopilot has been used to control the ship's heading angle by using a rudder. The rudder angle is controlled by the PID or adaptive mode which is built into the autopilot. The ship course-keeping control by the autopilot is of paramount importance since it is closely associated with navigational safety and economy. Any deviation from the original course not only may result in an increase in the sailing distance and thus ship operating costs but also is likely to cause serious maritime incidents like ship grounding and collision. In particular, sufficient steering capability should be ensured when a ship sails in rough seas. In this context, IMO (2017) has introduced the guideline pertaining to the course-keeping ability of a ship in adverse weather.

The direct CFD approach used in this paper has a strong advantage of estimating not only the combined seakeeping and manoeuvring performance but also the course keeping ability in various wave conditions. Recently, some researchers conducted studies on the course keeping control of the ONRT (Wang et al., 2017) and KCS (Kim et al., 2021) in waves, using the direct CFD method. However, these past studies only focused on the course keeping control in different wave directions. This section aims to provide a better understanding of the wavelength effect on the course keeping ability of a ship in waves.

Fig. 16 and Fig. 17 present the time histories of the rudder deflection angle, yaw angle, and yaw rate for each case when the ship was advancing forward based on the course-keeping control module. The signs of these values were determined in accordance with the coordinate systems established for modelling. For the rudder angle, a plus sign means the port rudder deflection which makes the ship turn to the port side, whilst a minus sign means the starboard rudder deflection which makes the ship turn to the starboard side. A positive value of the yaw angle and velocity indicates the direction of rotation to starboard, while

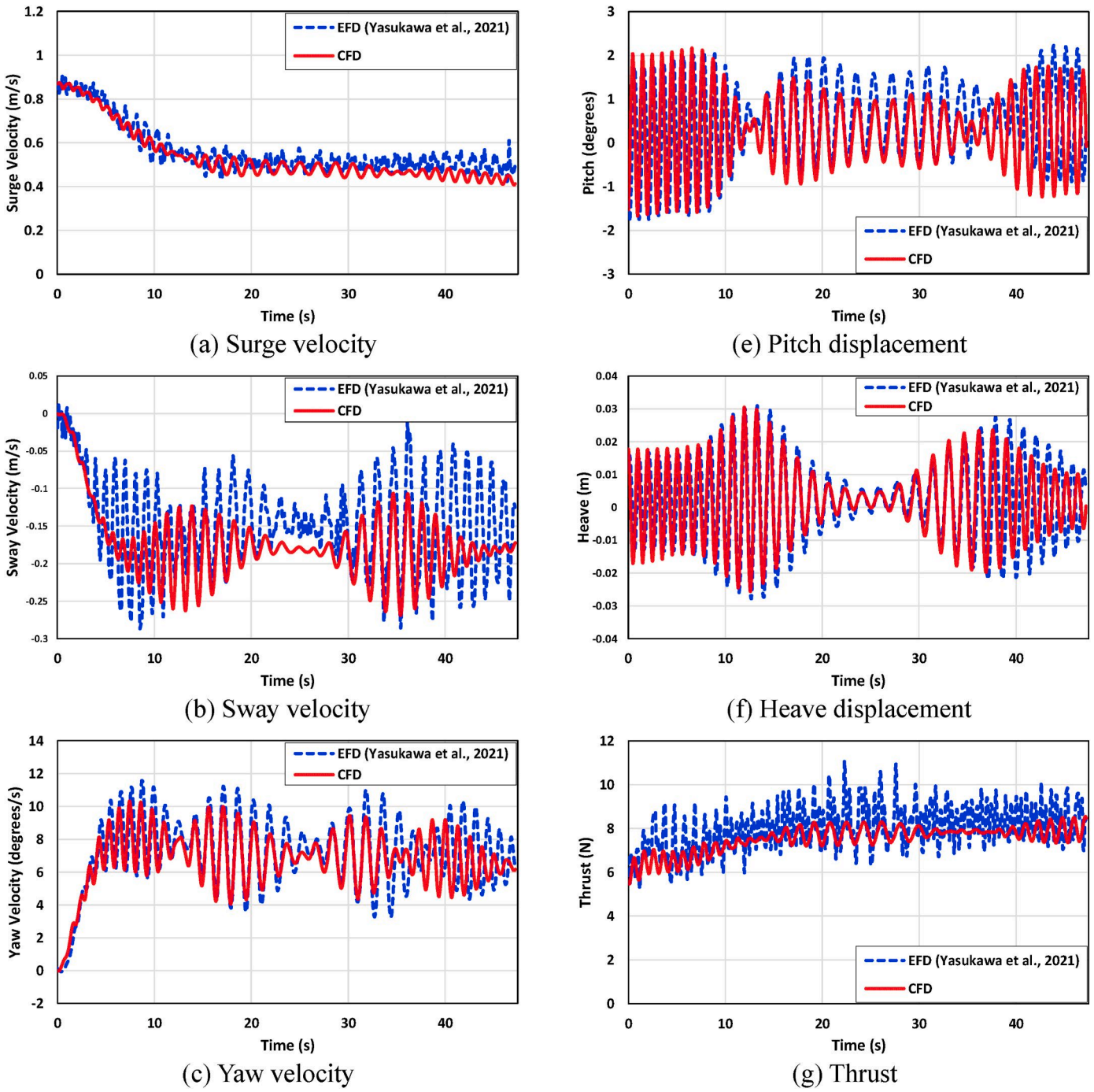
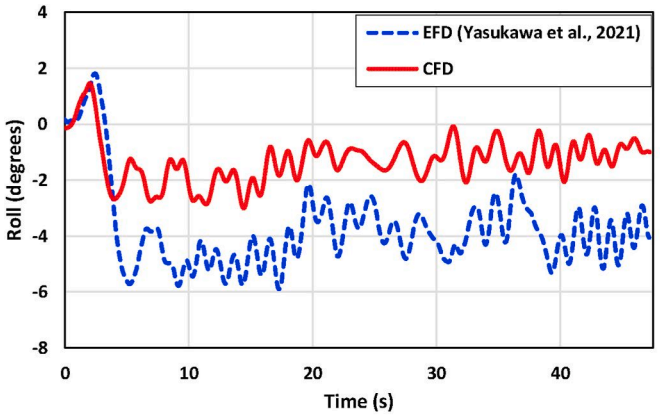


Fig. 13. The comparisons of the time histories of the ship velocities, motions, propeller characteristics, and turning trajectory during turning a circle manoeuvre in head wave, Kim et al. (2021).

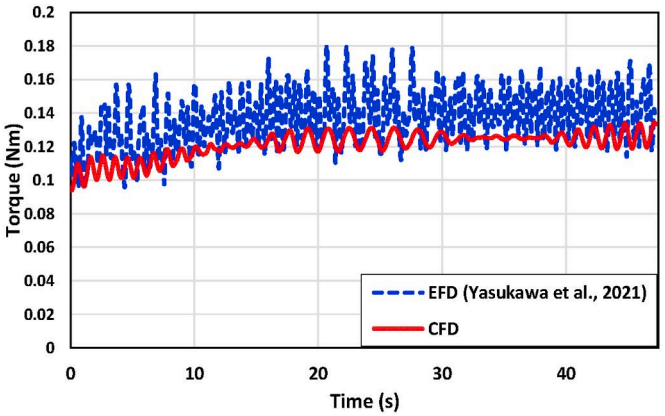
a negative value indicates the direction of rotation to port. It should be noted that the rudder angle amplitude and frequency can vary substantially depending on the gains of the PID controller, based on Eq. (8).

As shown in Fig. 16 (a), the rudder deflection was found to be very small under course keeping control in calm water since there are no environmental loads. The ship's heading angle was kept to be almost 0° with a maximum rudder deflection of 2.0°. The yaw velocity was also maintained to be almost 0°/s as seen in Fig. 17 (a). The small port rudder deflection may be closely related to a right-handed propeller effect which induces asymmetric pressure distribution acting on the rudder blade. Such an uneven pressure distribution generated a small yaw moment to move the bow to starboard. Thus, the rudder was deflected to the port side to control the ship's heading angle. It is of note that when the ship was moving forward at a constant speed, the symmetric wave profile around the hull was clearly observed, as seen in Fig. 15 (a). This would result in the even pressure distribution on both sides of the hull, which does not generate lateral force and yaw moments and consequently does not induce yaw deviation.

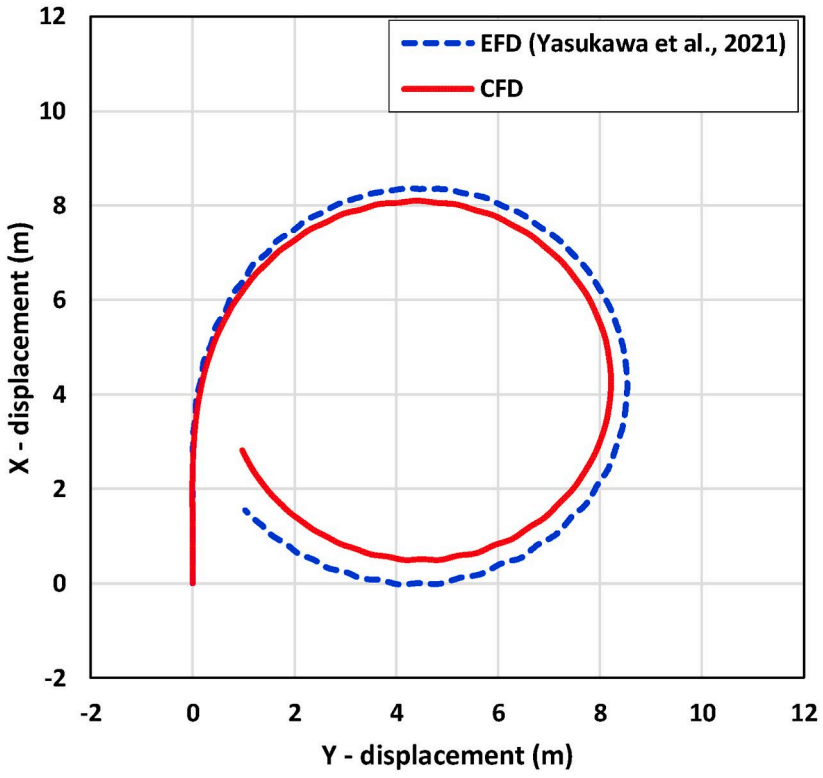
The amplitude of the rudder deflection became comparatively large in the bow waves, compared to the calm water condition. Similar maximum rudder deflections were estimated in all bow waves within a value of 7° as seen in Fig. 16 (b)–(f). As can be seen in these figures,



(d) Roll displacement



(h) Torque



(i) Trajectory in head waves

Fig. 13. (*continued*).

Table 6
Comparison of the main parameters of a standard turning circle manoeuvre in calm water, adapted from Kim et al. (2021).

Parameters	CFD	EFD (Yasukawa et al., 2021)	Error (%)
RPS at self-propulsion point	10.56	10.40	1.58
Advance (m)	9.30	9.29	0.01
Transfer (m)	3.95	4.16	4.86
Time for yaw 90° (s)	15.45	15.64	−1.22
Tactical diameter (m)	9.94	9.66	2.89
Time for yaw 180° (s)	31.50	30.50	3.28

Table 7
Comparison of the main parameters of a standard turning circle manoeuvre in head wave, adapted from Kim et al. (2021).

Parameters	CFD	EFD (Yasukawa et al., 2021)	Error (%)
RPS at self-propulsion point	13.38	13.2	1.36
Advance (m)	7.86	8.20	−4.11
Transfer (m)	3.09	3.35	−7.78
Time for yaw 90° (s)	12.52	13.29	−5.81
Tactical diameter (m)	7.97	8.37	−4.83
Time for yaw 180° (s)	23.74	25.32	−6.24

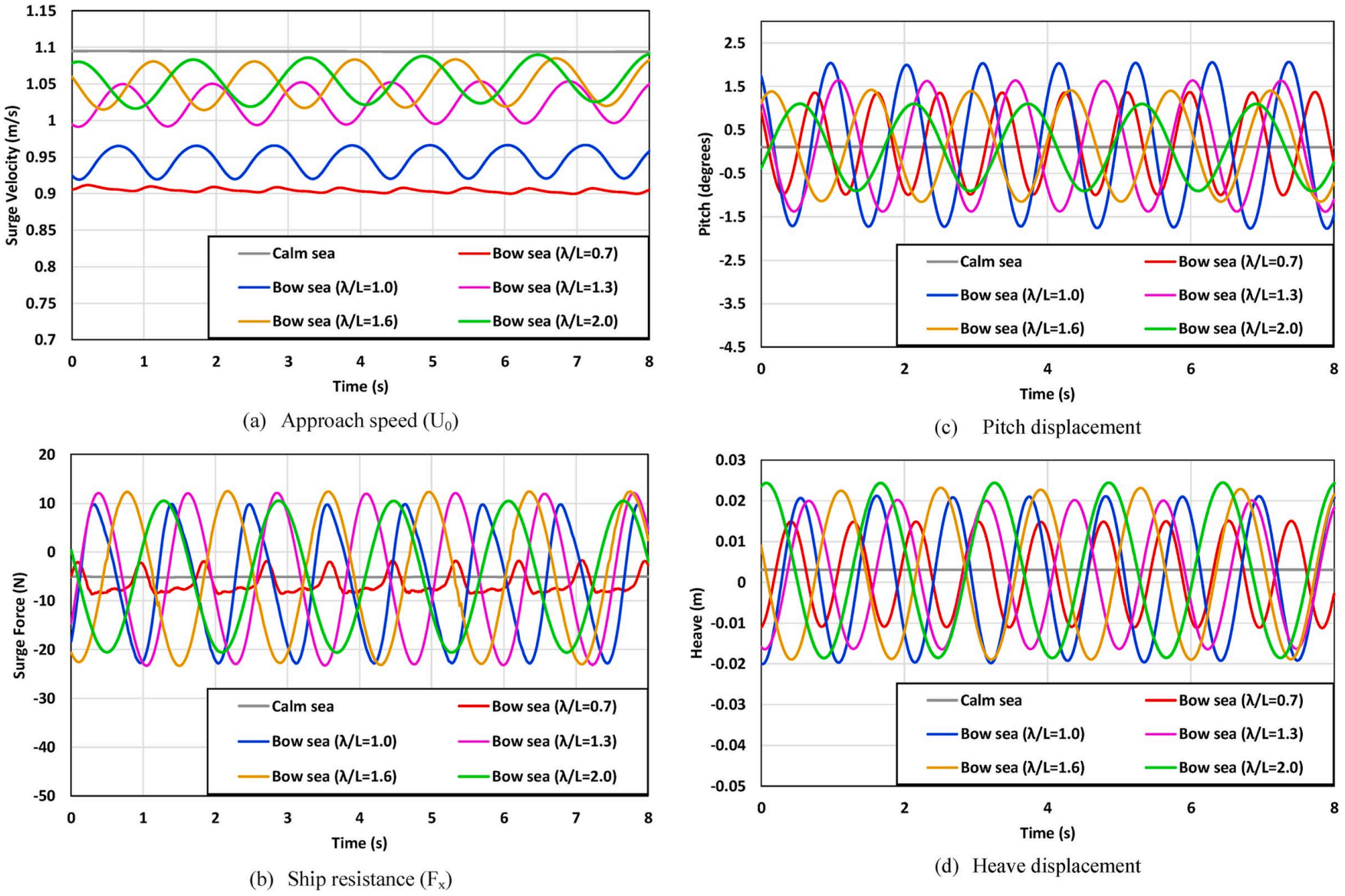
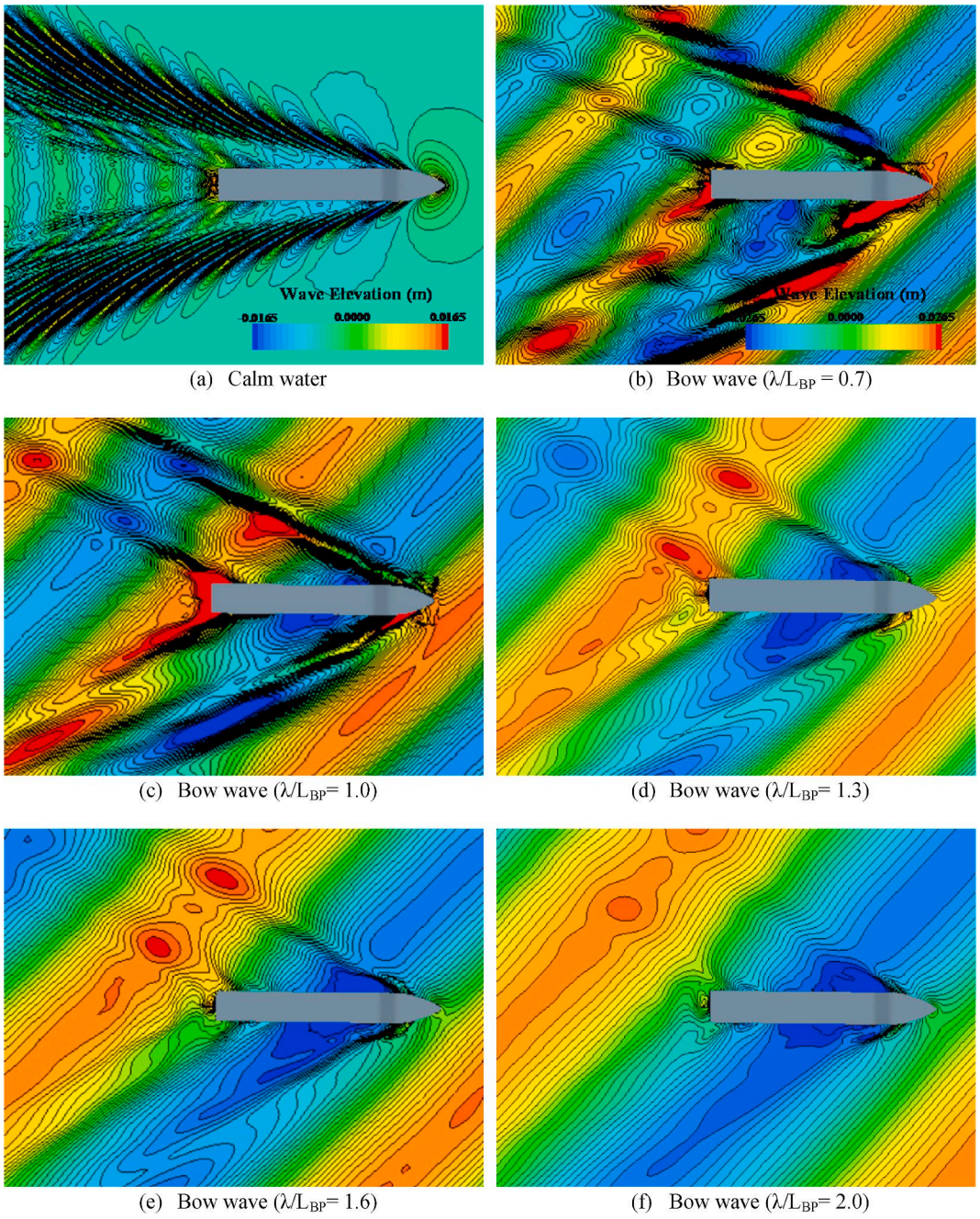


Fig. 14. Time histories of the approach speed, ship resistance, pitch motion and heave motion for all cases.

Table 8
Fourier series analysis of the approach speed, ship resistance, pitch motion and heave motion under course keeping control.

Case no. C	Approach speed U_0 (m/s)	Resistance F_x (N)		Pitch displacement (degree)		Heave displacement (m)	
	0th FS term	0th FS term	1st FS term	0th FS term	1st FS term	0th FS term	1st FS term
1 (Calm sea)	1.094	5.136	–	0.111	–	0.003	–
2 (Bow sea, $\lambda/L_{BP} = 0.7$)	0.904	6.470	2.082	0.175	1.139	0.002	0.013
3 (Bow sea, $\lambda/L_{BP} = 1.0$)	0.945	6.723	15.548	0.170	1.876	0.002	0.019
4 (Bow sea, $\lambda/L_{BP} = 1.3$)	1.024	4.930	17.029	0.094	1.499	0.001	0.018
5 (Bow sea, $\lambda/L_{BP} = 1.6$)	1.050	5.168	17.869	0.130	1.311	0.001	0.020
6 (Bow sea, $\lambda/L_{BP} = 2.0$)	1.054	5.102	15.477	0.118	0.999	0.003	0.022



(a) Calm water (b) Bow wave ($\lambda/L_{BP} = 0.7$)

(c) Bow wave ($\lambda/L_{BP} = 1.0$) (d) Bow wave ($\lambda/L_{BP} = 1.3$)

(e) Bow wave ($\lambda/L_{BP} = 1.6$) (f) Bow wave ($\lambda/L_{BP} = 2.0$)

Fig. 15. Free surface elevation under course keeping control before conducting the turning manoeuvre.

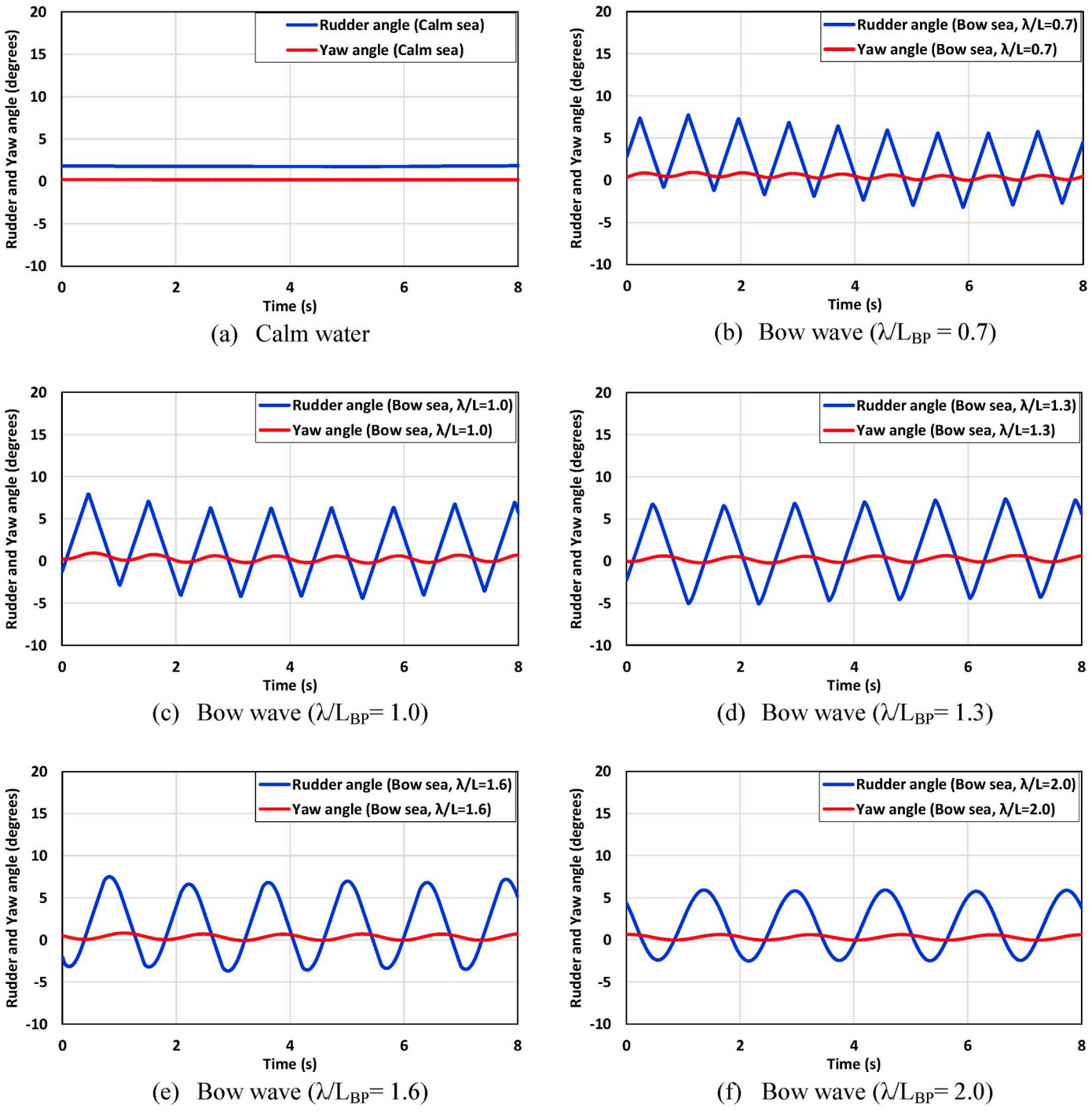


Fig. 16. Time histories of the rudder deflection and yaw angle under course keeping control in (a) calm sea, (b) bow sea ($\lambda/L_{BP} = 0.7$), (c) bow sea ($\lambda/L_{BP} = 1.0$), (d) bow sea ($\lambda/L_{BP} = 1.3$), (e) bow sea ($\lambda/L_{BP} = 1.6$), and (f) bow sea ($\lambda/L_{BP} = 2.0$).

rudder control is critical to maintain a desired heading of the ship in bow seas, regardless of the wavelength. The maximum yaw angle deviations were also observed to be about 1°, while the maximum yaw rates were predicted to be about 2°/s for all wave cases as seen in Fig. 17 (b)–(f). However, the steering frequency was found to differ depending on the wavelength, despite the same autopilot gains. The shorter wavelength was applied, the more frequent rudder deflection was observed. It is of note that the high steering frequency has been considered not to satisfy the requirements of navigation practice, as discussed in Yan et al. (2020). The authors stated that in real navigation practice, there exists a certain delay from the time when the rudder angle is given by the controller to the moment that the response of the steering engine is completed. Thus, it can be expected that the ship's heading control is more challenging in short waves in real operation. As presented in Fig. 15 (b)–(f), asymmetric wave profiles were clearly noticed in bow seas as opposed to calm sea, which may lead to a substantial lateral force and yaw moment with consequent changes in the ship's heading angle.

In order to visualise and explain the behaviour of the vessel under course keeping control, the snapshots of the pressure distribution patterns and wave elevations, and the time history of the ship's yaw moment were taken in a given period encounter and are shown in Fig. 18. The pictures are the snapshots from the simulation of Case 2. In

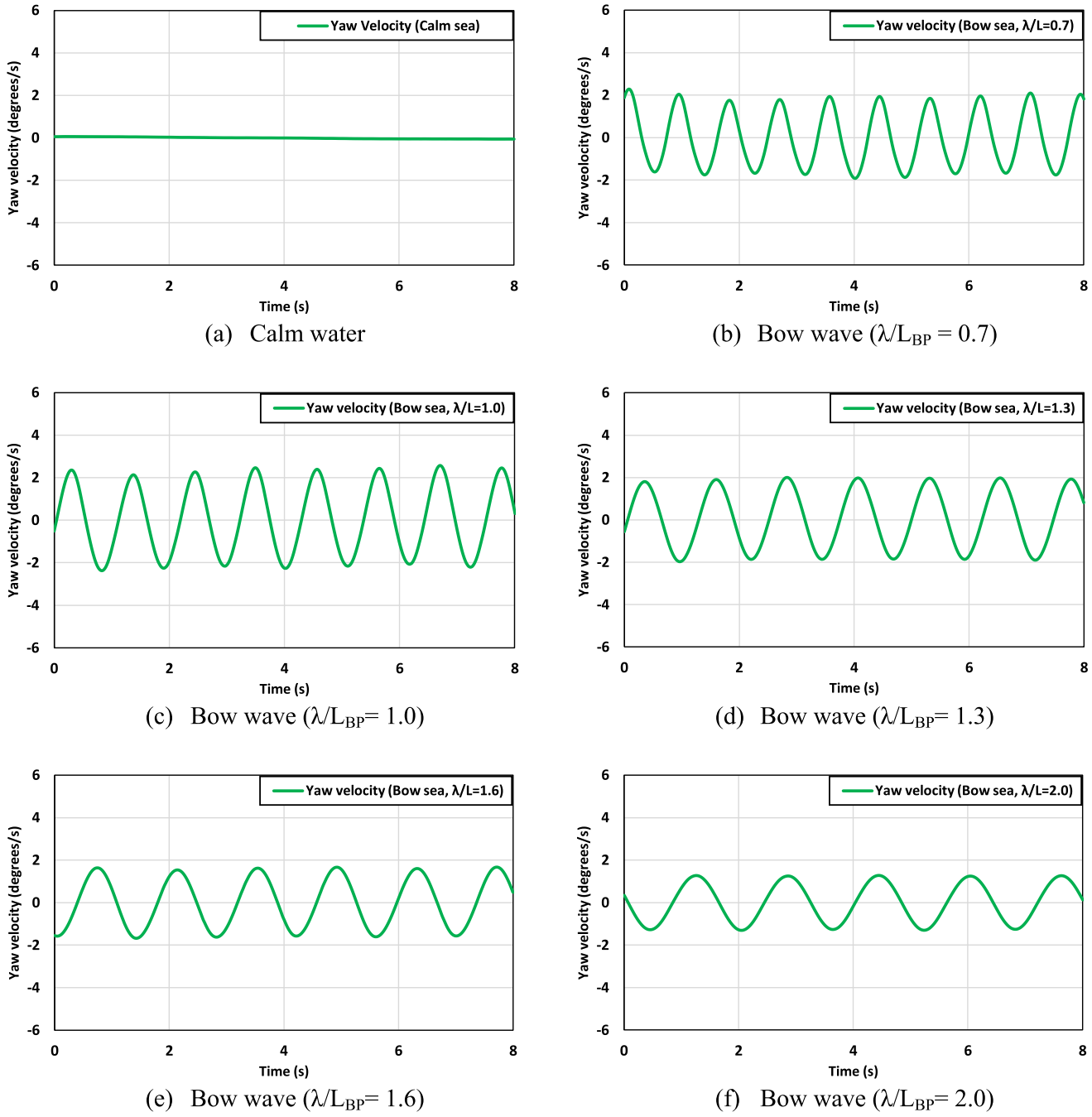
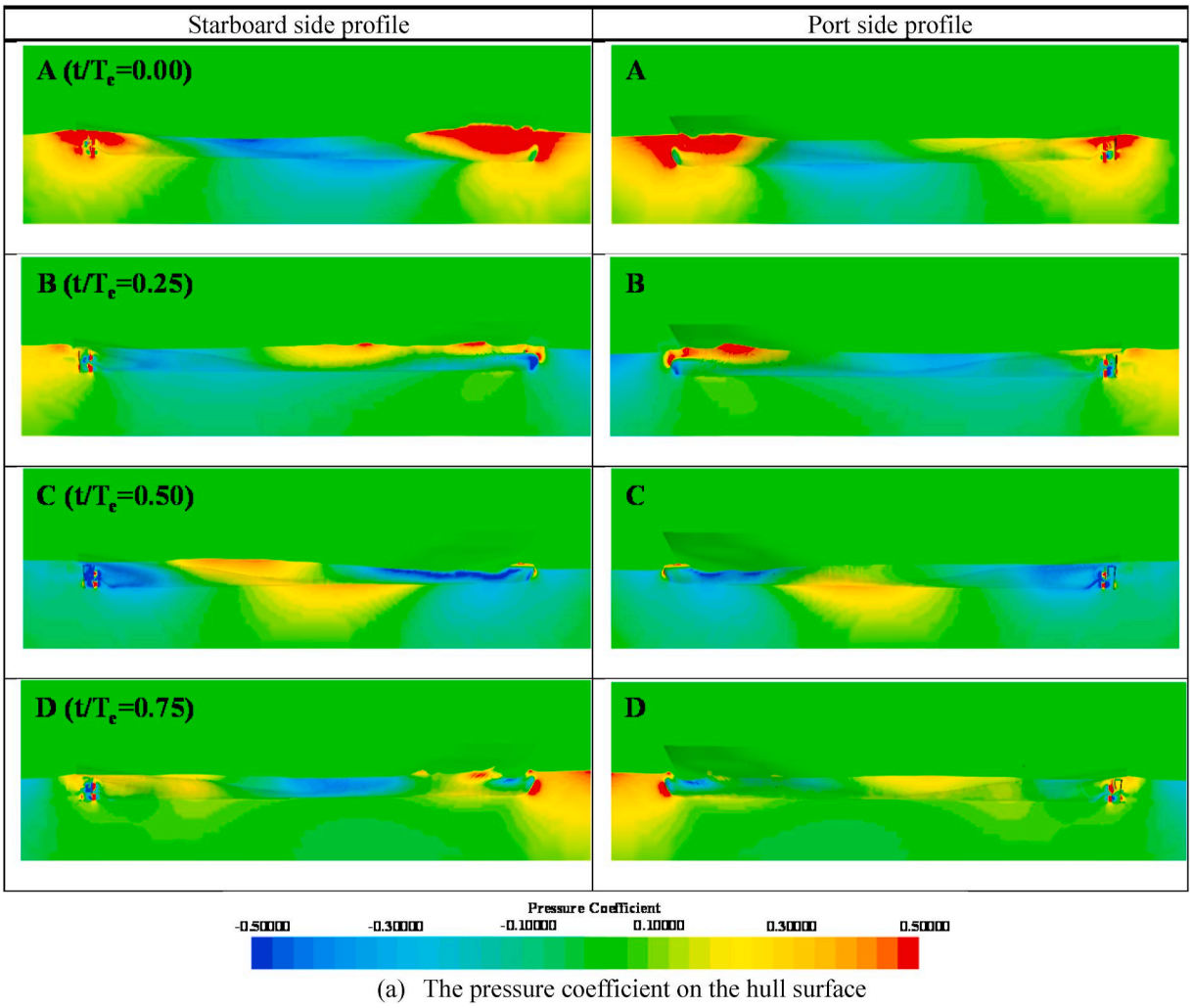


Fig. 17. Time histories of the yaw velocity under course keeping control in (a) calm sea, (b) bow sea ($\lambda/L_{BP} = 0.7$), (c) bow sea ($\lambda/L_{BP} = 1.0$), (d) bow sea ($\lambda/L_{BP} = 1.3$), (e) bow sea ($\lambda/L_{BP} = 1.6$), and (f) bow sea ($\lambda/L_{BP} = 2.0$).

the figure, the pressure was non-dimensionalised by dividing it by the dynamic pressure ($1/2\rho V_{ship}^2$). Phase A ($t/T_e = 0.00$) clearly shows that the surface pressure on the starboard bow was larger than that on the port bow when the wave crest passed on the ship's bow, as seen in Fig. 18 (a)–(b). The surface pressure on the port-stern side of the vessel was also observed to be larger than that on the starboard-stern side. Such a pressure distribution caused the maximum yaw moment to turn the ship to the port side, as shown in Fig. 18 (c). When the ship was positioned in Phase B ($t/T_e = 0.25$), the yaw moment acting on the ship was predicted at zero although the pressure difference between the starboard and the port side was clearly identified. This may be because the pressure difference existed near the centre of mass, which caused only the ship trajectory drift. When the wave crest subsequently passed on the midship of the ship (Phase C, $t/T_e = 0.50$), the pressure on the starboard-stern side was much larger than the port-stern side. In addition, the port-bow showed a larger surface pressure compared to the starboard-bow. The pressure distribution observed in Phase C led to the maximum yaw moment to turn the ship to the starboard side. Similar to Phase B, the surface pressure distribution on the hull in Phase D ($t/T_e = 0.75$) did not cause the yaw moment. The ship's heading angle was predicted to change depending on the pressure distribution on the hull by the oblique wave, while the rudder was deflected to control the yaw angle.

Fig. 19 presents the predicted ship trajectories under course keeping control for all the cases. The relatively large deviation from the original



(a) The pressure coefficient on the hull surface

Fig. 18. The snapshots of the pressure distribution patterns and free surface elevations, and the time history of the ship's yaw moment in a period of encounter under course keeping control.

course was observed in the short wave conditions ($\lambda/L_{BP} = 0.7$ and 1.0), whereas a very small deviation was noticed for the rest of the cases. This can be explained by the effect of the wave drift force on the ship, which was observed to be more significant in the shorter wave conditions. The more the number of encounter waves at a given period may lead to the stronger the impact of the wave drift force on the ship. As a result of this, the ship advancing forward in the short wave conditions was noted to be more drifted, as shown in Fig. 19. The ship trajectory drift directions were similar to the wave propagation direction. The predicted ship paths revealed that the deviation of the ship's course was dependent on the wavelength under oblique wave conditions. It was also estimated that the long wavelength conditions can cause the course of the ship to oscillate continuously near the planned route. It should be noted that the course keeping capability might be further improved when applying the optimum gains of the feedback controller with sufficient propulsion power.

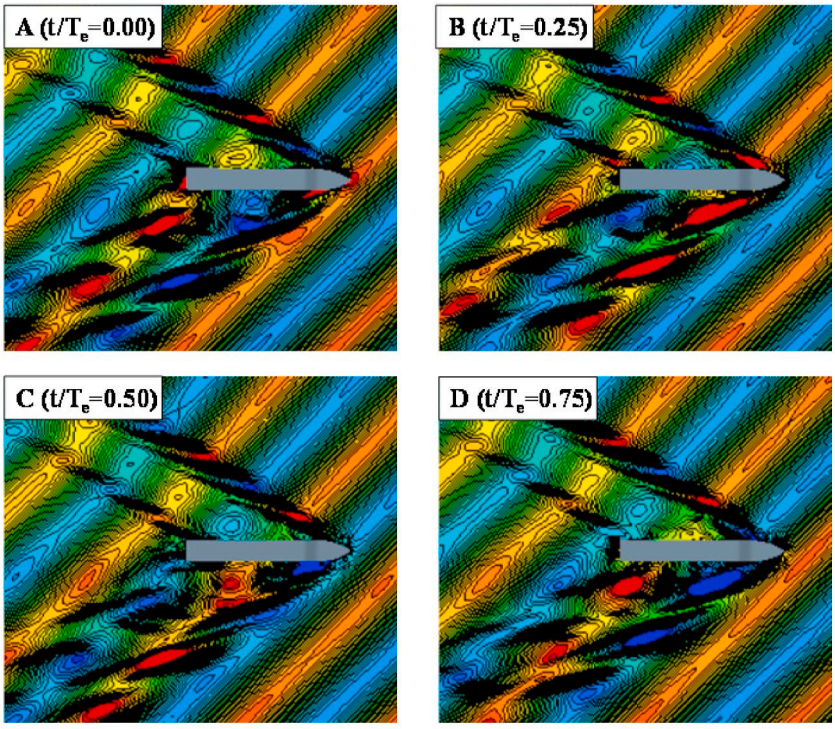
3.3. Turning circle manoeuvre

The turning manoeuvrability of a ship can be assessed through the standard turning circle manoeuvre. In this paper, the starboard turning tests were simulated to investigate the ship's turning ability. In the turning manoeuvre, the ship was moving forward at the approach speed and the rudder angle deflection towards 35° starboard was executed at maximum rudder rate. Subsequently, the ship started to respond turning in that direction. The ship model used for the turning circle manoeuvres was the same as that of the course keeping simulations with a propeller revolution of 13.38 RPS. Throughout all the simulations, the turning circle manoeuvres with only the heading angle variation up to 360° were performed based on the general procedure (IMO, 2002). According to the procedure, a 720° turn test is also recommended in order to fully evaluate environmental effects, but this requires a high computational time cost. Therefore, in this study, a 720° turn was not examined with the assumption that the 360° turn is sufficient to offer critical information about the manoeuvring performances of a ship in waves.

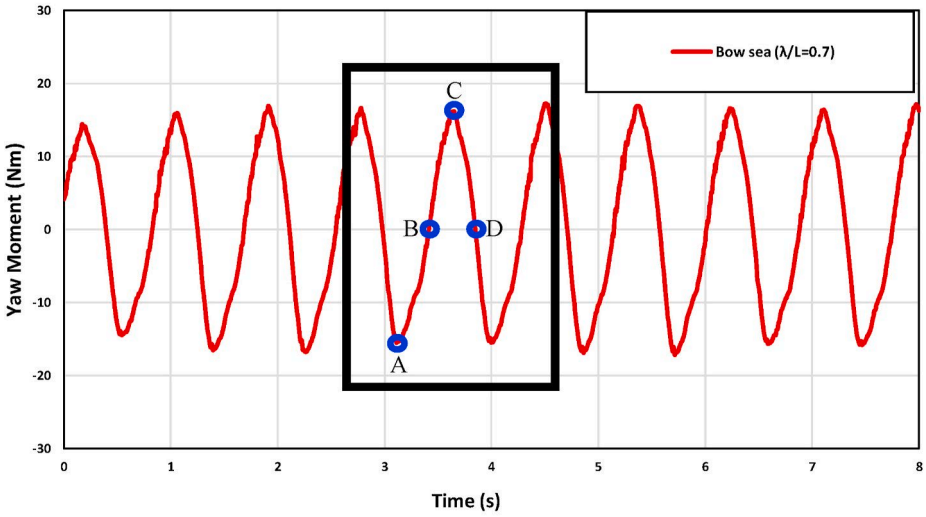
Turning circle manoeuvre can determine the advance, transfer, tactical diameter and the times associated with the ship completing these manoeuvres, which were identified by ITTC (2002). The measurements of these manoeuvring indices are considered as a means of quantifying the ship's inherent turning stability. The critical manoeuvring quantities are depicted in Fig. 20.

3.3.1. Time histories during the turning and turning indices

Fig. 21 illustrates the time histories of the predicted ship velocities, forces and moment during the standard turning manoeuvre. All values on the graphs are expressed in model-scale. Several black dash lines are depicted to represent the representative wave-encounter directions during the ship's turning to starboard. In order to execute the turning



(b) The free surface elevation around the ship



(c) The time history of the yaw moment acting on the hull

Fig. 18. (*continued*).

circle test, the rudder angle was deflected towards the starboard 35° during the moments when the wave trough passed on the midship of the model. It is of note that the timing of rudder deflection has little influence on the manoeuvring behaviours of a ship, as discussed in the study of Kim et al. (2019).

When the ship was sailing forward, the pressure distribution acting on the ship hull was found to vary depending on the wavelength of bow waves. This caused differences in the added resistance induced by the incident waves, and thus the ship experienced changes in the forward speed according to the wave condition. The propulsion characteristics were also influenced by the high-frequency wave-induced motions and the orbital wave-particle velocities (Liu et al., 2020). Such complex interactions between the ship and waves played a significant role in determining the behaviour of the turning manoeuvre.

In general, considerable fluctuations were observed in the ship velocities as well as the hydrodynamic forces and moments acting on the ship during the ship's turning because of the high-frequency wave-induced motions. As for the ship forward velocities as shown in Fig. 21 (a)–(f), they showed a gradual decrease after the rudder execution and finally converged to between 0.43 m/s and 0.5 m/s with some fluctuations around their mean value depending on the wavelength condition. This involuntary speed loss was caused by an increase in ship resistance,

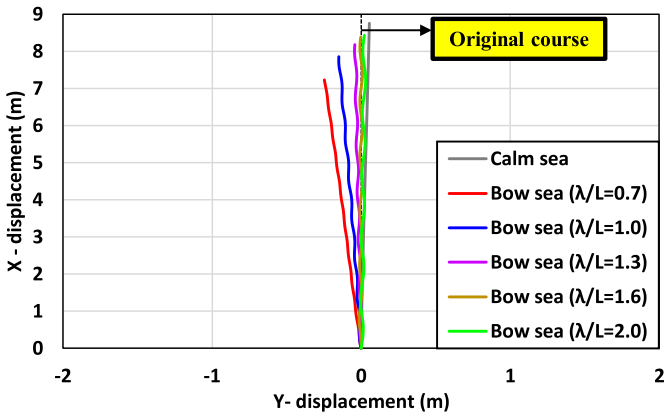


Fig. 19. Comparison of the predicted trajectories for all the cases.

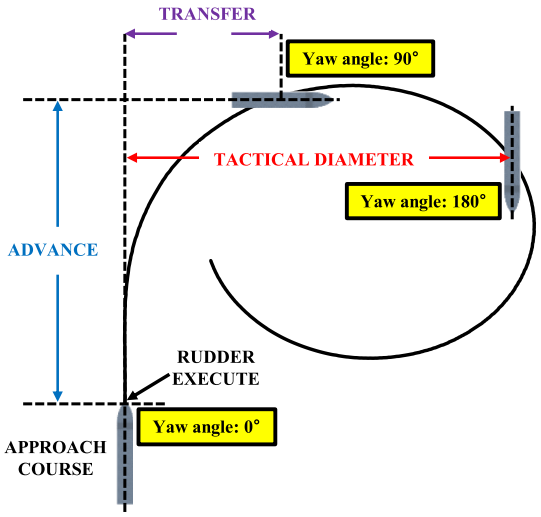


Fig. 20. The definitions of a turning circle.

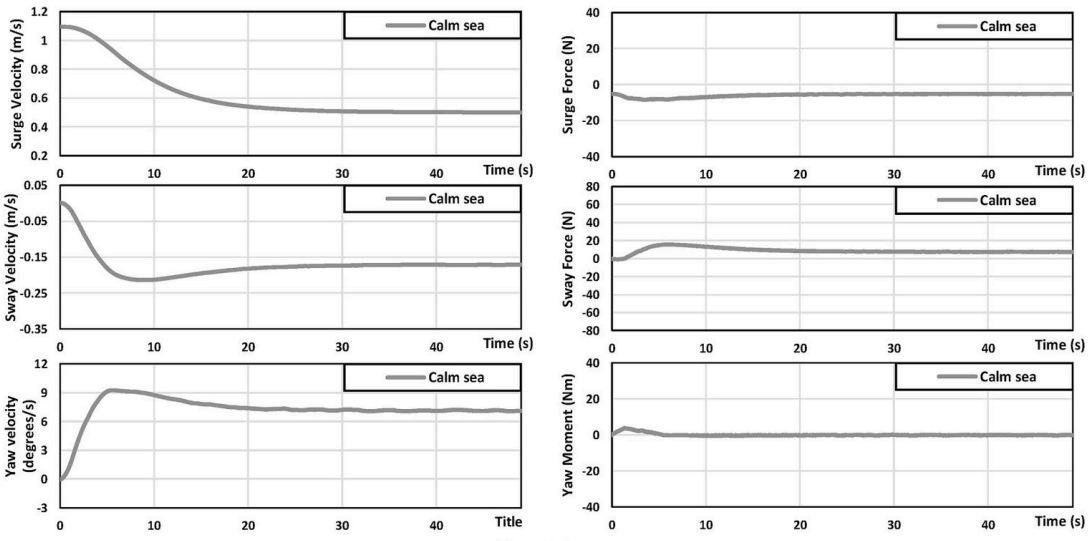
which is closely associated with a large drift angle while the ship is turning. The maximum speed loss for the calm water was noted approximately 30s after the rudder was deflected, whereas the maximum speed loss for wave cases occurred between about 16s–25s after the rudder execution. The speed loss rate between the initial approach speed and the steady mean value was estimated to be 54% for the calm water, 48% for the $\lambda/L_{BP} = 0.7$, 50% for the $\lambda/L_{BP} = 1.0$, 56% for the $\lambda/L_{BP} = 1.3$, 58% for the $\lambda/L_{BP} = 1.6$, and 56% for the $\lambda/L_{BP} = 2.0$. It was observed that the surge forces fluctuated significantly during the ship's turning manoeuvre under wave conditions, but the fluctuations almost disappeared at the moment when the ship experienced the port or starboard beam waves. Likewise, the surge velocities also showed similar trends.

For the sway velocities and forces, as shown in Fig. 21 (a)–(f), they tend to increase rapidly until approximately 8s after the rudder deflection to starboard and then showed a tendency to converge to similar values. The converged average values of the sway velocities were predicted to be −0.18 m/s for the calm water, −0.17 m/s for the $\lambda/L_{BP} = 0.7$, −0.18 m/s for the $\lambda/L_{BP} = 1.0$, −0.19 m/s for the $\lambda/L_{BP} = 1.3$, −0.2 m/s for the $\lambda/L_{BP} = 1.6$, and −0.19 m/s for the $\lambda/L_{BP} = 2.0$. The oscillations in the sway velocities and forces were observed to be larger when the ship experienced beam waves, whilst the relatively small oscillations were found under both the head and following seas. Regarding the yaw velocities and moments, the fluctuation magnitude varied depending on the wave-encounter direction. The largest fluctuations were clearly observed when the ship experienced the quartering waves. The maximum yaw rates occurred approximately 6s after the rudder execution. Then, yaw velocities gradually converged to the values which were predicted to be 7.31°/s for the calm water, 7.80°/s for the $\lambda/L_{BP} = 0.7$, 7.87°/s for the $\lambda/L_{BP} = 1.0$, 8.20°/s for the $\lambda/L_{BP} = 1.3$, 8.33°/s for the $\lambda/L_{BP} = 1.6$, and 8.05°/s for the $\lambda/L_{BP} = 2.0$.

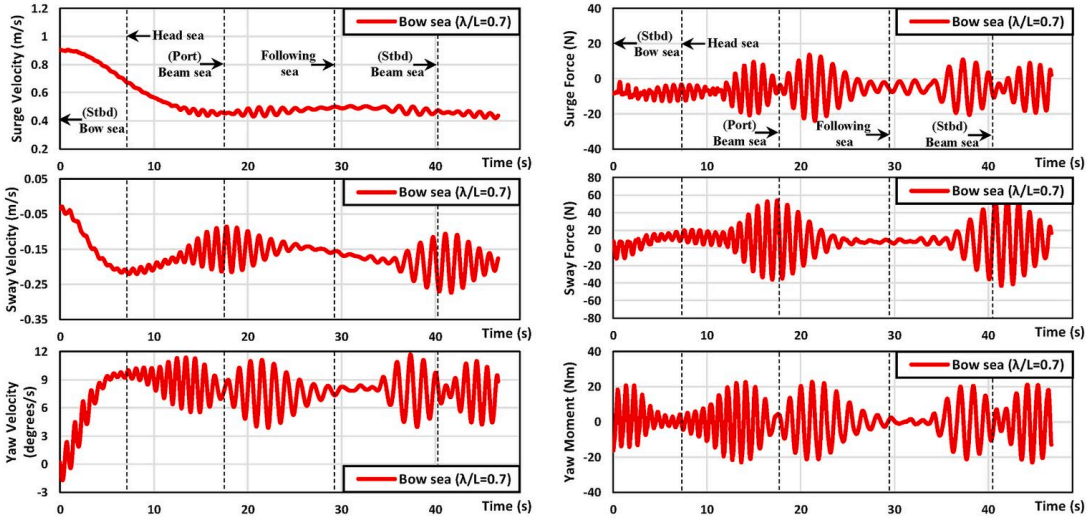
Table 9 shows the comparison of the turning manoeuvring quantities associated with the ship's turning ability. In real navigation, the advance, transfer, and tactical diameter have been recognised as the most critical manoeuvring quantities to understand the ship's manoeuvring performance at sea. These indices were closely correlated to the ship's horizontal motions during the turning circle manoeuvre. Accordingly, the hydrodynamic forces and moment acting on the ship in the horizontal plane were identified as the main factors involved in the critical manoeuvring parameters. These forces and moments on ship turning in waves resulted from complex interactions between the rudder, propeller, hull, and waves. In particular, the imposed wavelength conditions had a significant effect on the hydrodynamic forces and moment during the turning manoeuvre. The turning circle trajectories were found to differ depending on the wavelength. As in ship advance and transfer, the ship velocities in the horizontal plane (surge, sway, and yaw velocities) played an important role in determining these distances. In general, the greater forward speed and the longer time to turn by 90°, the larger the advance and transfer can be. The maximum advance and transfer distances were observed in cam water (case 1) due to the relatively large forward speed and the long 90° turning time. Although the 90° turn time was relatively long, the ship experienced the minimum advance and transfer in the bow waves of $\lambda/L_{BP} = 0.7$ (case 2) because the forward speed was the lowest at the beginning of the turning manoeuvre. Interestingly, despite the relatively small forward speed, the advance and transfer for the $\lambda/L_{BP} = 1.0$ (case 3) were similar to those observed for the $\lambda/L_{BP} = 1.3$ and 1.6 (case 4 and 5) due to the longer time to turn by 90°. It was noted that the $\lambda/L_{BP} = 1.3$ and 1.6 cases caused a larger yaw moment than that for the $\lambda/L_{BP} = 1.0$ as the ship started turning in waves. This accelerated the ship's yaw velocities, and 90° turns were achieved much earlier. In the same manner, the ship advancing in the bow waves of $\lambda/L_{BP} = 1.3$ and 1.6 experienced larger sway velocities than the $\lambda/L_{BP} = 1.0$ case. Among the wave cases, the maximum advance and transfer were predicted in the bow waves of $\lambda/L_{BP} = 2.0$ (case 6) due to the relatively large ship's speeds. As seen in Fig. 21, the $\lambda/L_{BP} = 1.3$, 1.6, and 2.0 conditions caused a large lateral force and yaw moment in the early stage of the turning, and hence large sway and yaw velocities were observed. The tactical diameters were also clearly found dependent on the wavelength. It was revealed that the relatively large tactical diameters were observed in the bow waves of $\lambda/L_{BP} = 1.3$, 1.6, and 2.0 under which large lateral force and yaw moment were observed at the early stage of the turning.

IMO (2002) states that the ship's turning ability is considered satisfactory if the ship advance does not exceed 4.5 ship lengths (L_{BP}) and the tactical diameter does not exceed 5.0 ship lengths in the turning circle test. As seen in Fig. 22, all the manoeuvring quantities were assessed by the IMO criteria. It was revealed from this comparison that the turning ability of the KCS in both calm water and waves can be considered satisfactory.

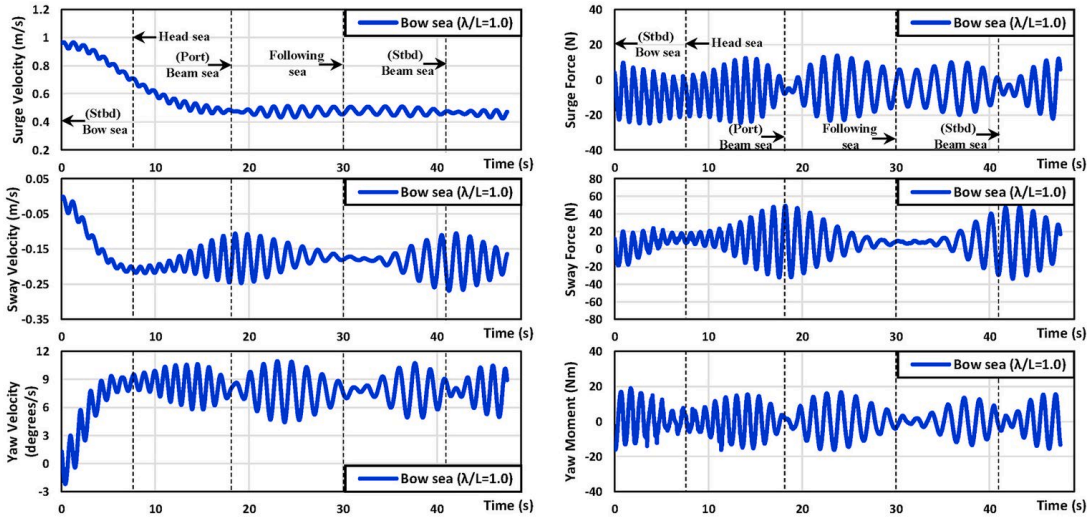
Fig. 23 describes the comparison of the predicted turning trajectories of the KCS for all the cases. For the correct comparisons of the trajectories, all the rudder execution points were shifted to the origin point (0,0). Generally, it was found that the size of the turning circle was found to have a proportional relationship to the wavelength. It is of note that very similar trajectories can be found in the bow waves of $\lambda/L_{BP} = 1.0$, 1.3, and 1.6, like the turning indices. The smallest turning circle of the KCS was observed in the bow waves of $\lambda/L_{BP} = 0.7$, which may be attributed to the shortest wavelength. In addition, the instantaneous free surface elevation around the ship hull during the starboard turning manoeuvre in bow waves is presented in Fig. 24. It is of note that the generated Kelvin wave by the manoeuvring ship was obviously more visible in shorter waves.



(a) Calm water

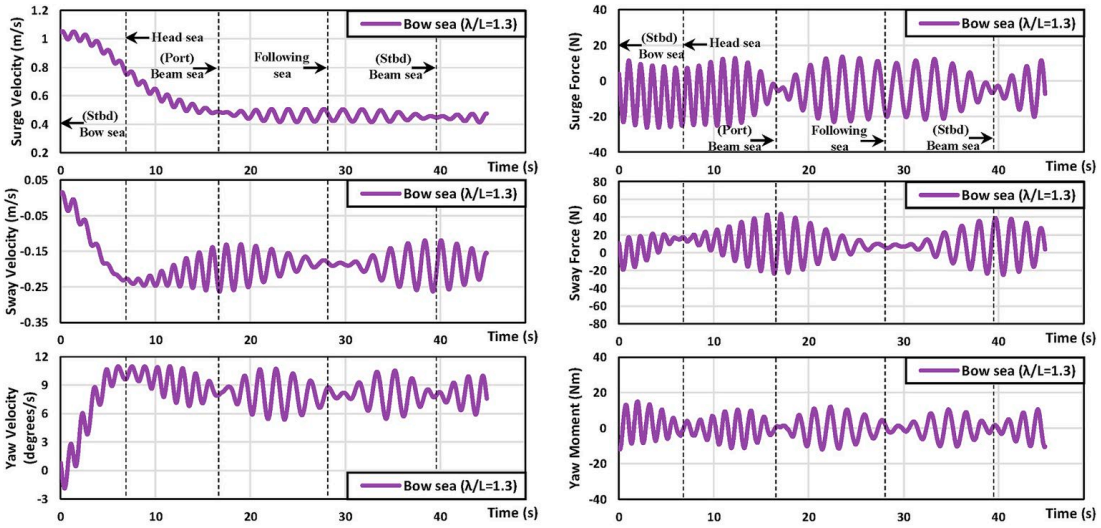


(b) Bow wave (λ/L_{BP} = 0.7)

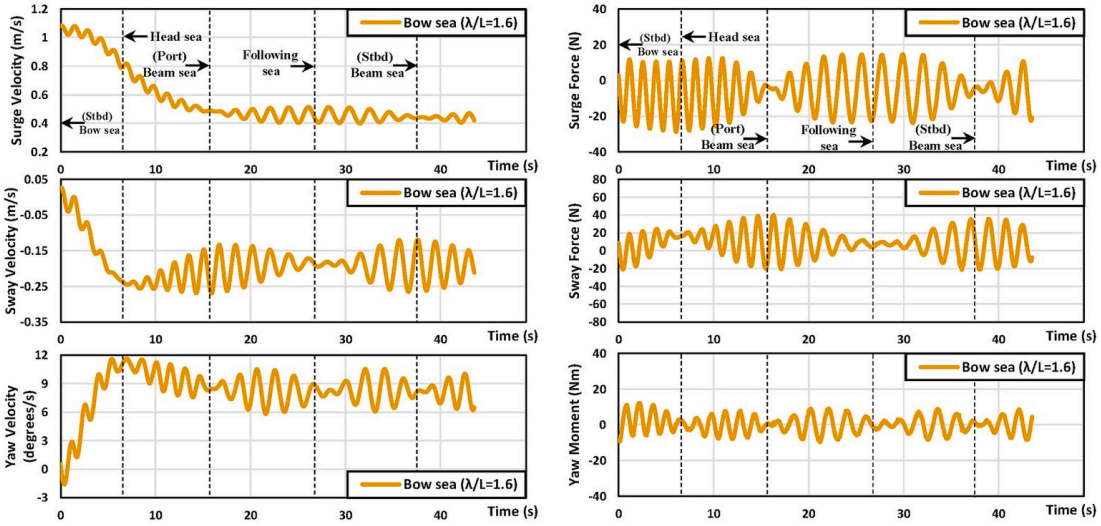


(c) Bow wave (λ/L_{BP}= 1.0)

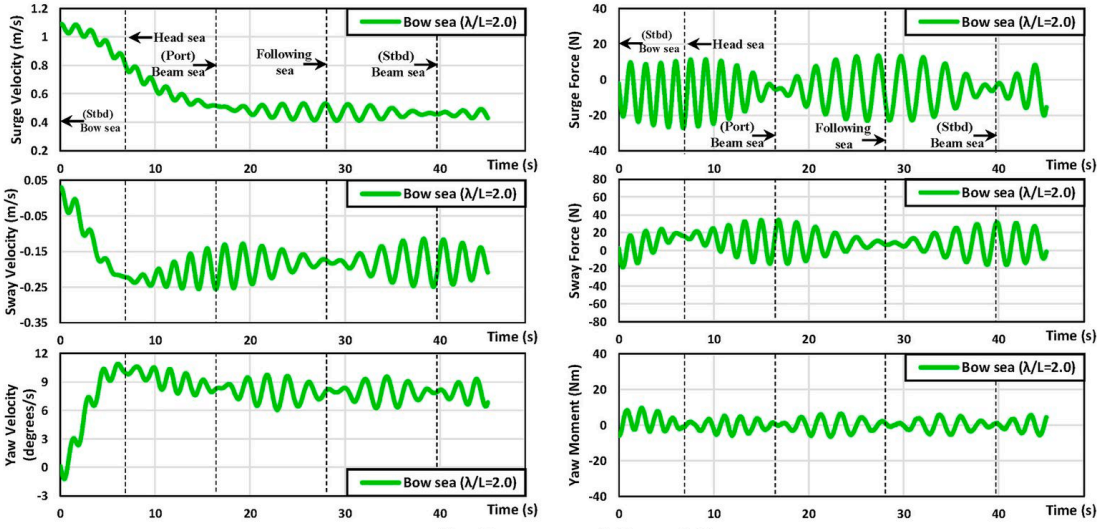
Fig. 21. Time histories of the ship velocities, forces, and moment during a turning manoeuvre in waves.



(d) Bow wave (λ/L_{BP}= 1.3)



(e) Bow wave (λ/L_{BP}= 1.6)



(f) Bow wave (λ/L_{BP}= 2.0)

Fig. 21. (*continued*).

Table 9
CFD results: turning indices in calm water and regular waves.

Parameters (CFD results)	Calm sea (case 1)	Bow sea (λ/L_{BP}= 0.7) (case 2)	Bow sea (λ/L_{BP}= 1.0) (case 3)	Bow sea (λ/L_{BP}= 1.3) (case 4)	Bow sea (λ/L_{BP}= 1.6) (case 5)	Bow sea (λ/L_{BP}= 2.0) (case 6)
Advance (m)	9.55	7.96	8.83	8.81	8.71	8.94
	(3.13L_{BP})	**(2.60L_{BP})**	**(2.89L_{BP})**	**(2.88L_{BP})**	**(2.85L_{BP})**	**(2.92L_{BP})**
Transfer (m)	4.07	2.69	3.09	3.02	3.01	3.36
	(1.33L_{BP})	**(0.88L_{BP})**	**(1.01L_{BP})**	**(0.99L_{BP})**	**(0.98L_{BP})**	**(1.09L_{BP})**
Time for yaw 90 degrees (s)	12.31	12.21	13.08	11.71	11.24	11.59
Tactical diameter (m)	9.82	7.03	7.82	7.74	7.60	8.23
	(3.21L_{BP})	**(2.29L_{BP})**	**(2.56L_{BP})**	**(2.53L_{BP})**	**(2.49L_{BP})**	**(2.69L_{BP})**
Time for yaw 180 degrees (s)	24.20	23.35	24.38	22.49	21.56	22.20

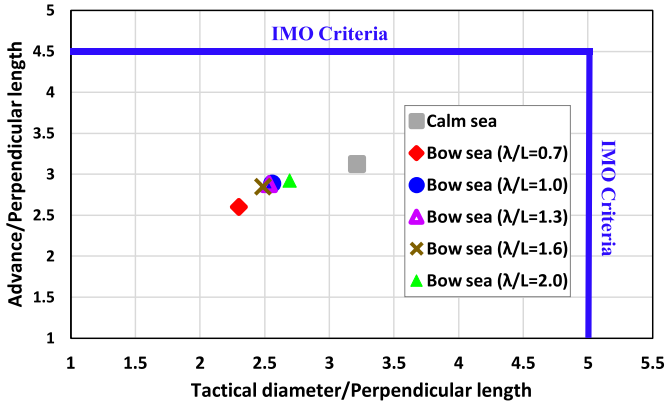


Fig. 22. Comparison of the predicted turning indices in waves with IMO turning criteria.

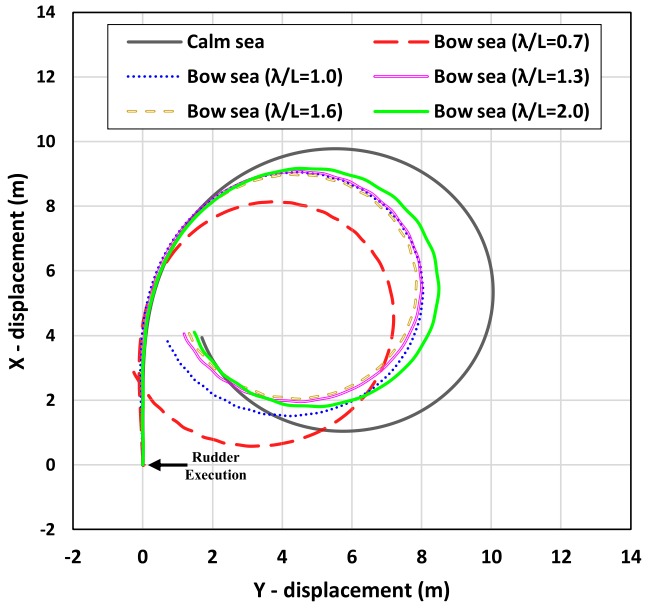


Fig. 23. Comparison of the predicted turning circle trajectories.

3.3.2. *Wave-induced motions during turning manoeuvre*

Fig. 25 shows the time histories of ship motions, i.e., pitch, heave, and roll as well as relevant forces and moments during the standard turning manoeuvre. Unlike the calm water case, the ship motions, forces, and moments were observed to experience considerable oscillations during the ship's turning in waves, which may be attributed to the high-frequency wave-induced motions. In addition, such frequencies continued to change during the ship's turning due to instantaneous variations in the ship's velocity and wave-encounter direction. For example, a ship advancing in the bow waves (case 2–6) encountered the starboard bow wave (0° turn), the head wave (45° turn), the port beam wave (135° turn), following wave (225° turn), starboard beam wave (315° turn), and starboard bow wave (360° turn) in series after starting the starboard turning manoeuvre. The ship's turning has contributed to continual changes in the wave-encounter frequency, consequently the ship motion frequency in waves.

It can be clearly seen from Fig. 25 (a) that the pitch motion and moment in calm water converged to almost zero value without any fluctuations during the turning manoeuvre. The heave motion and force in calm water also followed a similar trend. On the other hand, it was revealed from Fig. 25 (b)–(f) that the waves have a great influence on the ship's motions, forces, and moments during the ship's turning. While the ship was turning, the maximum pitch motion and moment were found when the ship experienced the head or bow waves whereas the minimum values were noted in beam waves. As for the heave motion and force, the maximum amplitudes were predicted when encountering the beam waves, while the minimum values were observed under the following sea conditions. Such variations in the ship motions, forces, and moments were observed more clearly in shorter waves, shown in Fig. 25 (b)–(c).

As discussed in Hasanvand and Hajivand (2019), the roll motion was observed to be significantly affected by the turning manoeuvre (Fig. 25 (a)–(f)). When the rudder was deflected to the hard-over angle (35°), the forces acting on the rudder blade caused the ship to heel to the starboard (to the centre of the turning circle). These initial roll angles were estimated in all the cases within a value of 2°. Subsequently, the ship was heeled to the port side (to the outside) due to the hydrodynamic forces and the centrifugal force acting on the hull. The maximum roll angles were noted to be 5.7° for the calm water, 4.4° for the $\lambda/L_{BP} = 0.7$, 4.4° for the $\lambda/L_{BP} = 1.0$, 4.7° for the $\lambda/L_{BP} = 1.3$, 5.4° for the $\lambda/L_{BP} = 1.6$, and 6.0° for the $\lambda/L_{BP} = 2.0$. The ship's roll angles finally converged to between 1 and 2° with some fluctuations around their average value. It is of note that the roll motion can vary depending on several factors such as the rudder angle, the approach speed, and the loading condition.

4. Conclusion and discussion

This research has demonstrated the effectiveness of the direct CFD model to predict the ship's manoeuvrability under various wave conditions. As coupled with the 5th-order Stokes wave model and wave forcing capability, this model was successful in providing an understanding of the manoeuvring behaviour of the KCS in different wavelength conditions. With the provision of time history data regarding the critical manoeuvring quantities, the simulation results were able to directly indicate the correlations between the ship's manoeuvrability and wavelength, which confirmed the impact of the wavelength on the manoeuvring performance of the ship in regular waves. The key findings of this research can be highlighted as follows:

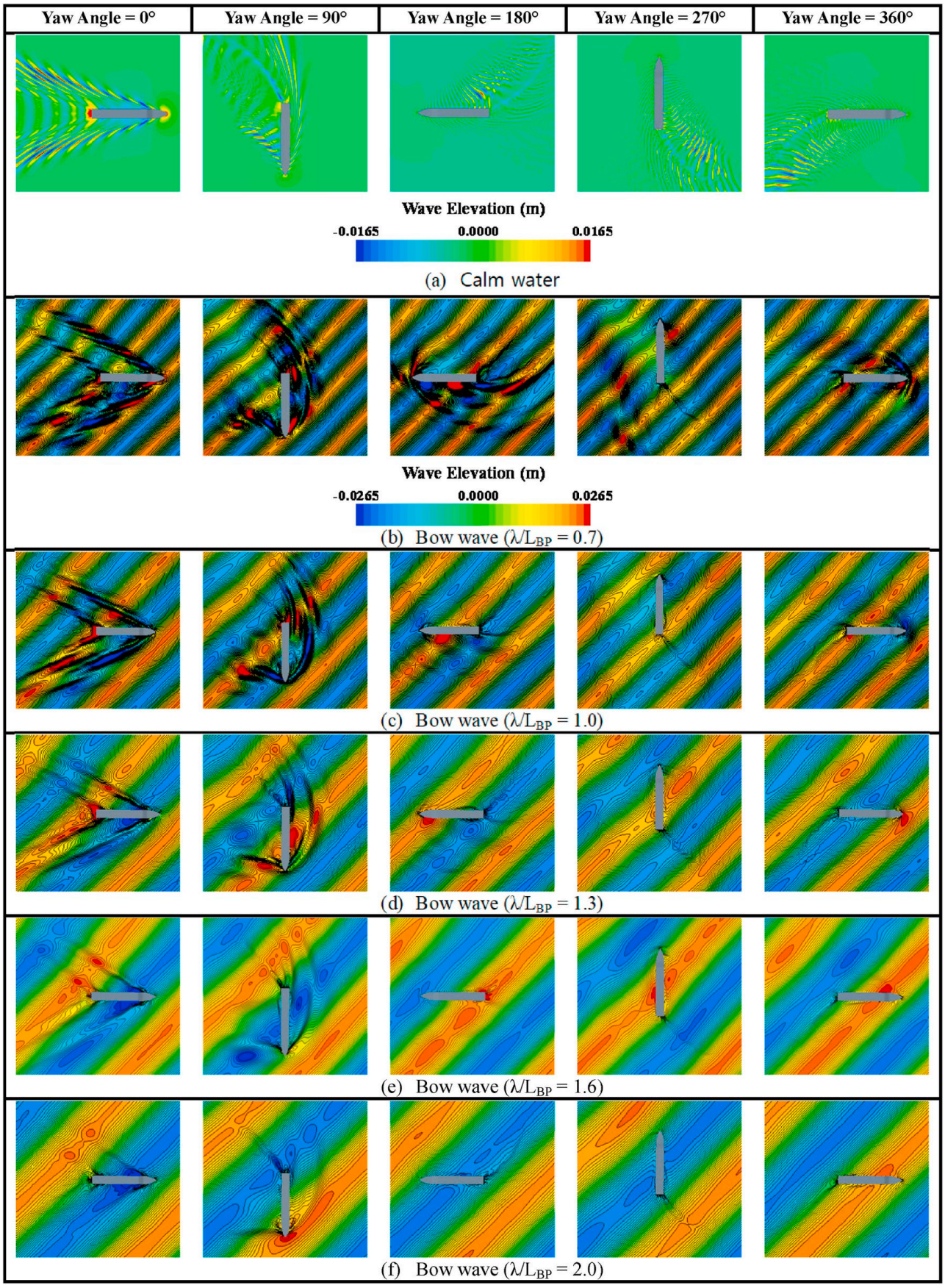
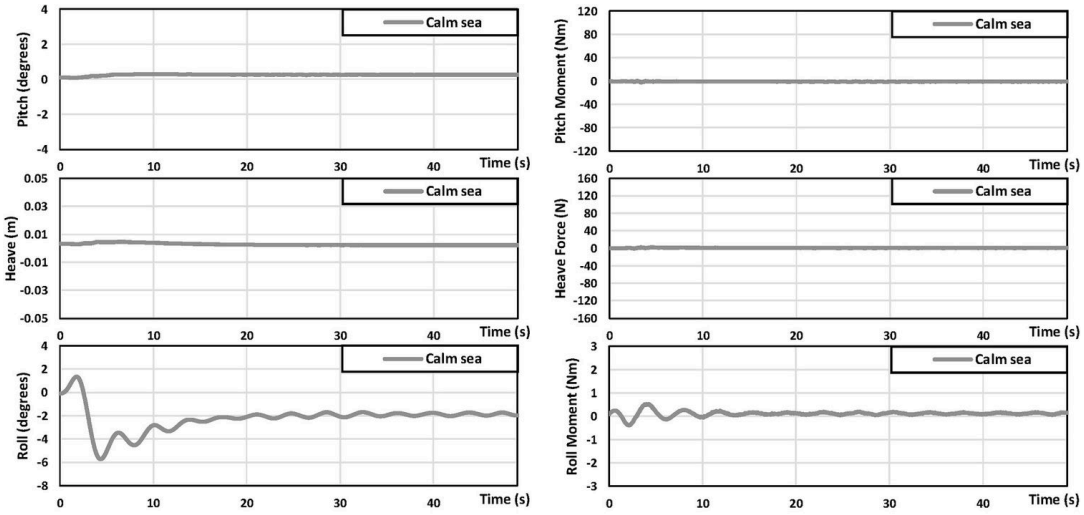
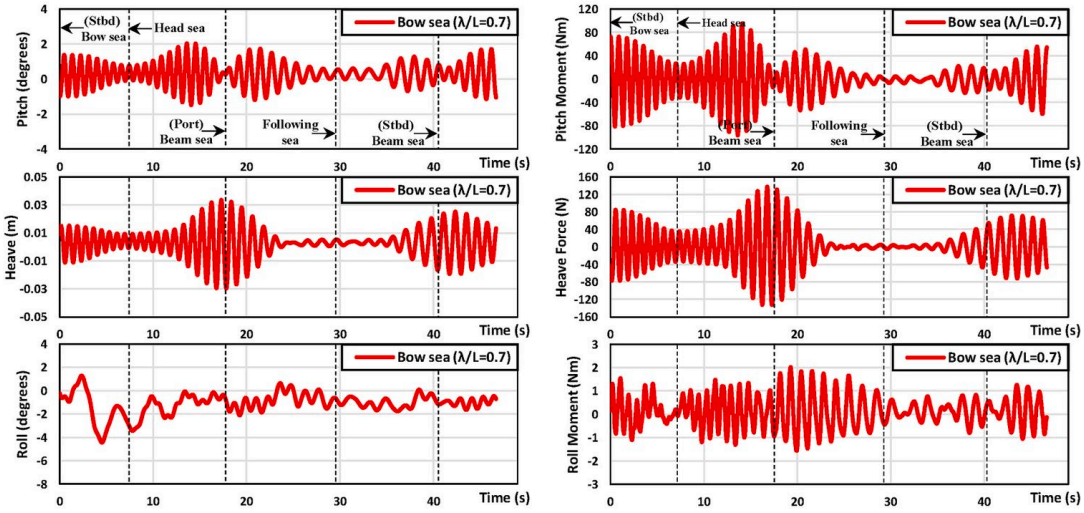


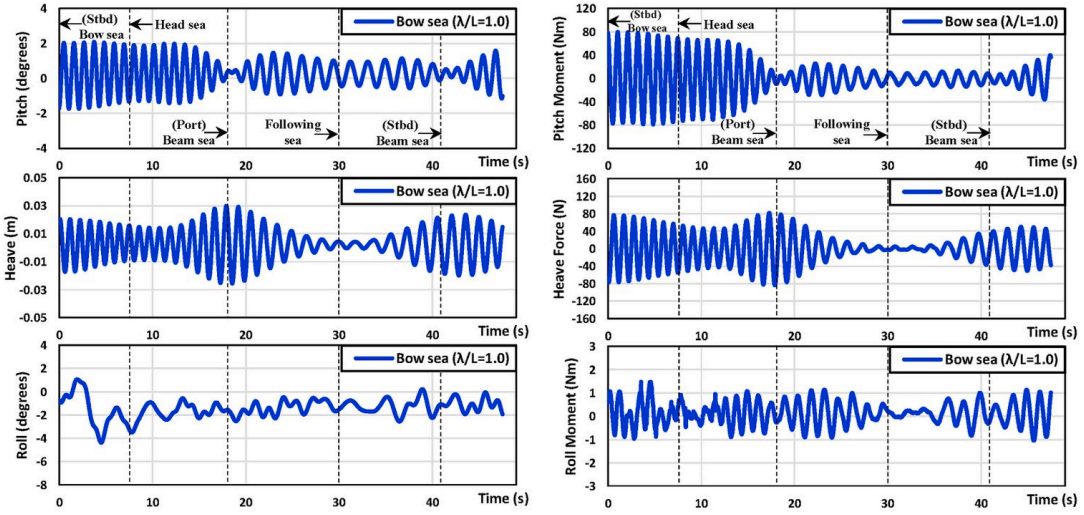
Fig. 24. The free surface elevation during the standard turning circle manoeuvre in waves.



(a) Calm water

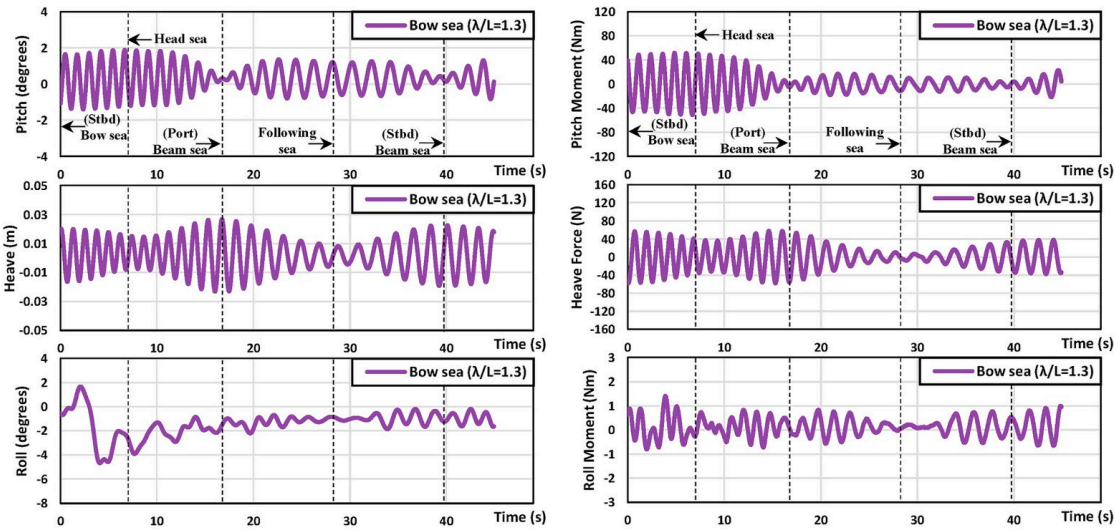


(b) Bow wave ($\lambda/L_{BP} = 0.7$)

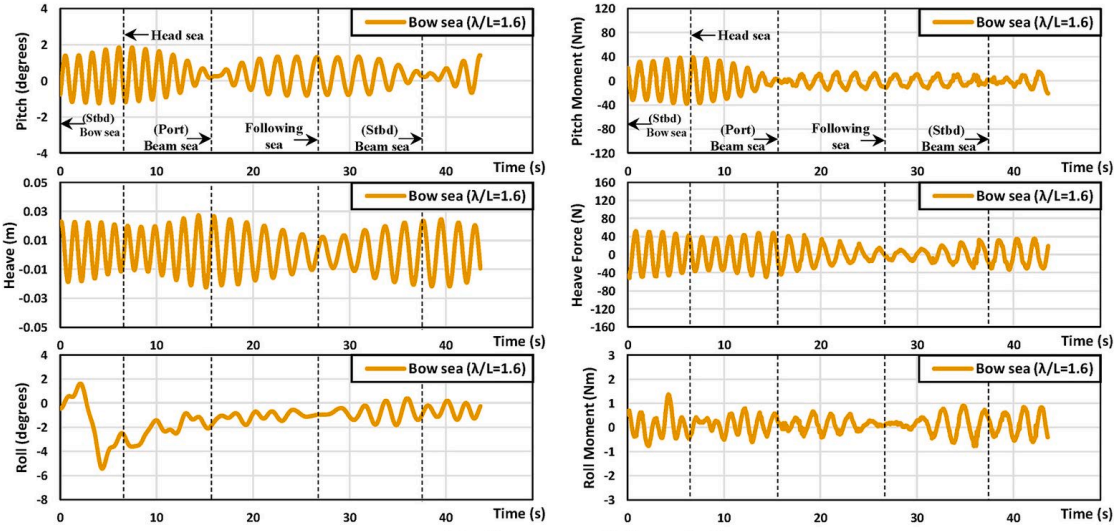


(c) Bow wave (λ/L_{BP}= 1.0)

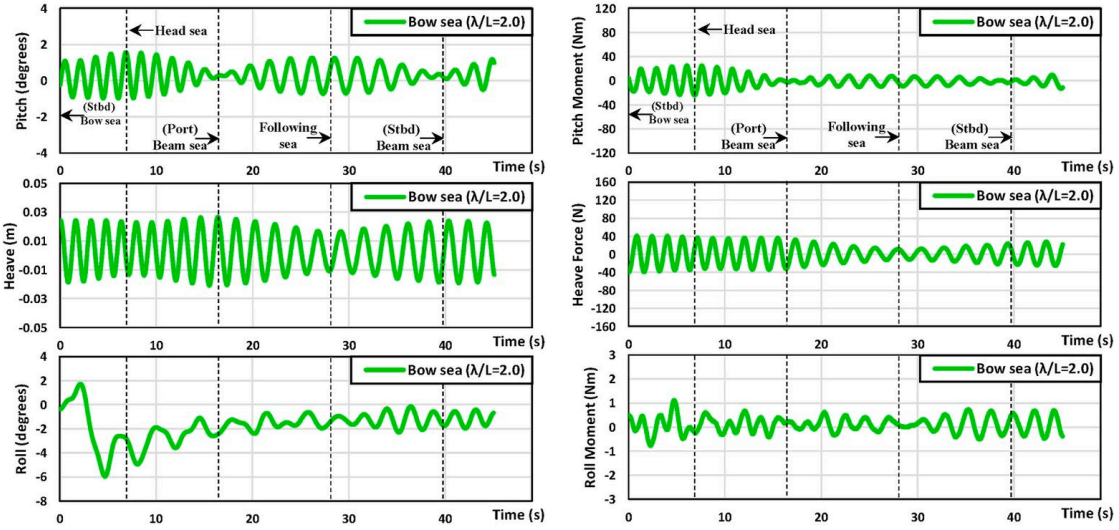
Fig. 25. Time histories of ship motions, forces and moments acting on the ship during turning manoeuvres.



(d) Bow wave (λ/L_{BP}= 1.3)



(e) Bow wave (λ/L_{BP}= 1.6)



(f) Bow wave (λ/L_{BP}= 2.0)

Fig. 25. (*continued*).

1) For the course keeping manoeuvres in waves, the approach speed was found to have a strong correlation with the wavelength. The approach speed was noted to increase with the increase in the wavelength at a given wavelength range ($0.7 < \lambda/L_{BP} < 2.0$). The minimum approach speed was observed to be 0.904 m/s for the $\lambda/L_{BP} = 0.7$, whilst the maximum approach speed was noted to be 1.054 m/s for the $\lambda/L_{BP} = 2.0$. The seakeeping motions also differed depending on wavelength conditions. The largest oscillation amplitude was identified in the bow waves of $\lambda/L_{BP} = 1.0$, whereas the largest heave motion was found in the bow waves of $\lambda/L_{BP} = 2.0$. As for the course keeping capabilities, simulation results revealed that the rudder control was a critical element to keep the ship straight in all bow wave conditions. Similar large rudder deflections were predicted approximately 7° in all bow waves, regardless of the wavelength. However, the steering frequency was found to vary significantly depending on the wavelength. The frequency of the rudder deflection was confirmed to have an inverse relation to the wavelength.
2) For the standard turning circle manoeuvres in waves, it was identified that the critical manoeuvring turning quantities were strongly affected by the wavelength and thus lead to significant changes in the turning trajectories. The minimum value of the advance, transfer, and tactical diameter was estimated under the shortest wavelength of $\lambda/L_{BP} = 0.7$ with a minimum turning circle radius. Interestingly, similar manoeuvring indices and trajectories were estimated in the bow waves of $\lambda/L_{BP} = 1.0$, 1.3, and 1.6. The longest wavelength of $\lambda/L_{BP} = 2.0$ caused the maximum increase of the manoeuvrability parameters which are the closest to the quantities of the calm water condition.

Although the wavelength was found a crucial parameter on the manoeuvring behaviour of the KCS, in practice, navigation officers have only access to the shipyard data which are generally measured by full-scale sea trials or model-scale experiments. Unfortunately, such data offer limited information on the ship manoeuvrability under the calm water conditions; not able to predict a wide range of real sea conditions. For this reason, this research will act as a valuable resource to provide a practical insight into understanding the manoeuvrability of commercial vessels. For example, the obtained manoeuvring information from this research can support navigators in proper decision-making for manoeuvring actions in waves to avoid ship-ship collision, especially when executing a large alteration of course with the maximum rudder deflection. Hence, it will be helpful for reducing marine accidents caused by improper knowledge of wave effects on ship operation.

A study of Tezdogan et al. (2015) revealed that the wave height can lead to substantial changes in the ship resistance and ship motions. It is highly believed that wave height is also likely to affect ship manoeuvrability as wavelength does. In this context, this research could further add values by investigating the effects of the wave height on the ship's manoeuvrability in waves as a next step. A proper understanding of the impacts of both wavelength and height will make a significant contribution to enhancing navigational safety at sea.

CRediT authorship contribution statement

Daejeong Kim: Writing – original draft, Formal analysis, Investigation, Validation, Visualization, Data curation. **Soonseok Song:** Writing – review & editing, Formal analysis, Investigation. **Byongug Jeong:** Writing – review & editing, Formal analysis, Investigation. **Tahsin Tezdogan:** Conceptualization, Methodology, Supervision, Writing – review & editing, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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